

Supplementary Materials:

AI vs. Humans: Does AI Face Harsher Moral Judgements? A Meta-Analysis¹

Author's Note:

Correspondence can be addressed to Olivia Hill-Cousins, oh298@exeter.ac.uk.

¹Olivia Hill-Cousins, University of Exeter. ORCID: <https://orcid.org/0009-0009-4436-8812>

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1 Materials to Supplement/Enhance Main Manuscript

1.1 Power Sensitivity Curves

As noted in the main manuscript, the power-sensitivity analyses were used to determine the minimum detectable effect size (MDES) for each model. These MDES values were then used as the equivalence bounds for the TOST procedure, ensuring that the bounds reflected the actual precision afforded by the obtained sample.

Each curve shows the estimated power across a range of effect sizes for a given iteration. The square marker indicates the point at which the curve reaches 80% power. The dotted line projects this point onto the x-axis, and the accompanying label reports the corresponding effect size.

Figure S1 presents the power curves for the full and trimmed datasets. Figure S2 shows the curves for the moderator analyses. For continuous moderators, one curve is shown per moderator; for categorical moderators, one curve is shown for each level.

Figure S1

Power Sensitivity Curves of Main Meta-Analytical Models

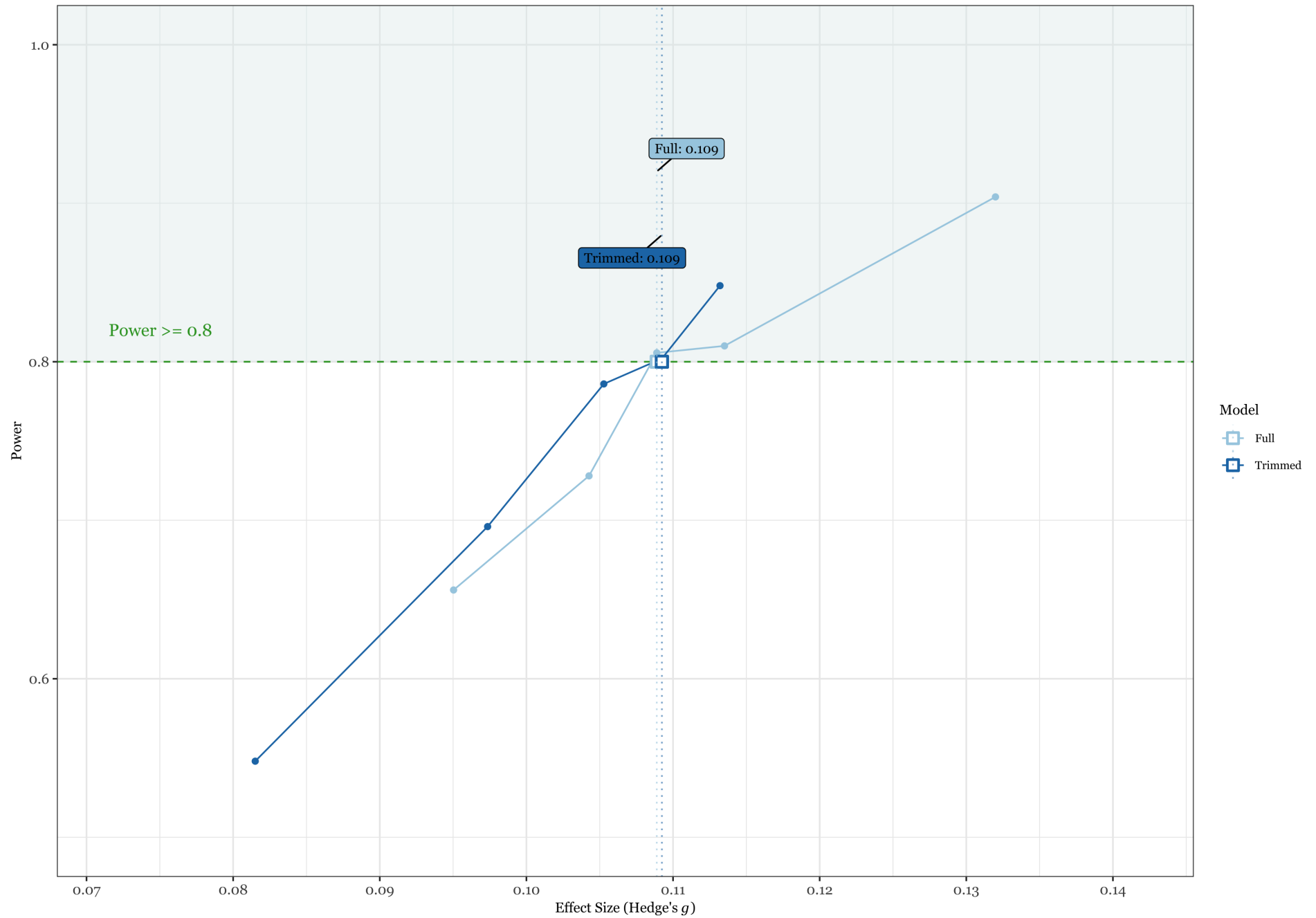
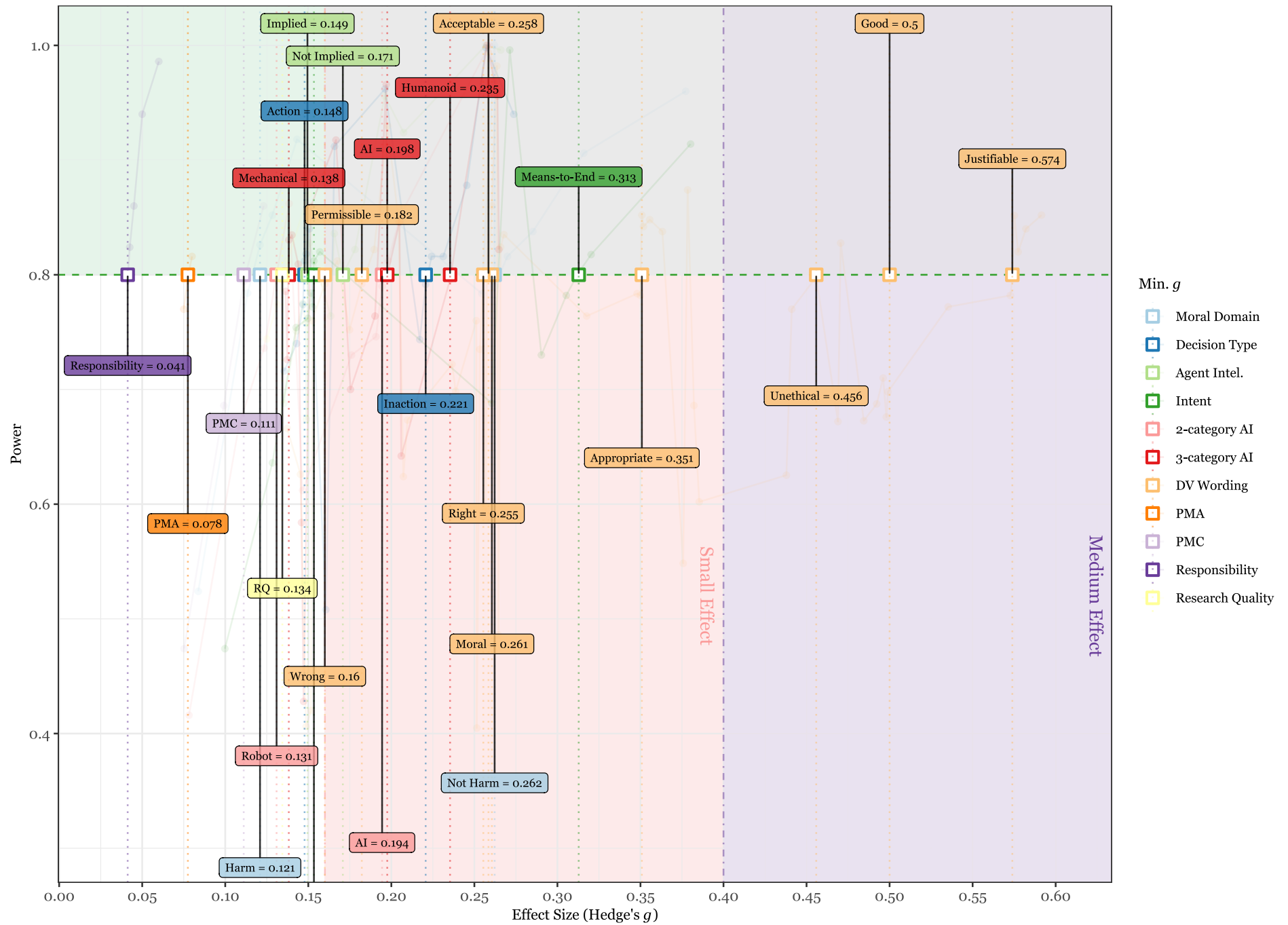


Figure S2

Power Sensitivity Curves of the Moderation Analytical Models



1.1.1 TOST for Full & Trimmed Datasets with MDES as Bounds

The bounds for TOST in the manuscript were set to the SESOI, as pre-registered, for the main/overall analyses. Below are the TOST results for both models with bounds set to their respective MDES values instead.

Full Model: $g = -0.07$, 90% CI $[-0.14, -0.01]$, bounds ± 0.11 , $p = 0.170$

Trimmed Model: $g = -0.08$, 90% CI $[-0.15, -0.02]$, bounds ± 0.11 , $p = 0.268$

1.2 Outliers

As noted in the main manuscript, the trimmed dataset includes only those outliers that were also identified as influential. In contrast, Table S1 lists all effect sizes flagged as outliers by the conservative detection method described in the main manuscript, regardless of whether they were influential.

Table S1

Table of all outliers flagged by the conservative method and their respective statistics for the influential diagnostics.

Article	Study	Ps ID	g	Cooks		DFBETAS		Hat	
				Value	Flag	Value	Flag	Value	Flag
Banks et al., (2021)	1	1	0.09	0.00		0.00		0.00	
		1	0.09	0.00		0.00		0.00	
		1	0.02	0.00		0.00		0.00	
		1	0.02	0.00		0.00		0.00	
	2	2	0.03	0.00		0.01		0.00	
		2	0.03	0.00		0.01		0.00	
		2	0.03	0.00		0.01		0.00	
		2	0.03	0.00		0.01		0.00	
		2	-0.20	0.00		-0.01		0.00	
		2	-0.20	0.00		-0.01		0.00	
		2	-0.20	0.00		-0.01		0.00	
		2	-0.20	0.00		-0.01		0.00	
Bigman et al., (2018)	3	3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.57	0.00		0.00		0.00	
		3	-0.51	0.00		0.00		0.00	
		3	-0.51	0.00		0.00		0.00	
		3	-0.51	0.00		0.00		0.00	

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Article	Study	Ps ID	g	Cooks		DFBETAS		Hat	
				Value	Flag	Value	Flag	Value	Flag
Cassioli et al., (2023)	1	5	-0.52	0.00		0.00		0.00	
		5	-0.52	0.00		0.00		0.00	
		5	-0.52	0.00		0.00		0.00	
		5	-0.52	0.00		0.00		0.00	
		5	-0.52	0.00		0.00		0.00	
		5	-0.52	0.00		0.00		0.00	
		5	-0.78	0.00		0.00		0.00	
		5	-0.78	0.00		0.00		0.00	
		5	-0.78	0.00		0.00		0.00	
		5	-0.78	0.00		0.00		0.00	
		5	-0.78	0.00		0.00		0.00	
		5	-0.78	0.00		0.00		0.00	
		6	-1.08	0.03		-0.16		0.01	
		7	-0.08	0.00		-0.01		0.00	
		8	0.05	0.00		0.00		0.00	
Chu et al., (2023)	2	9	0.51	0.00		0.03		0.00	
		7	0.01	0.00		0.01		0.00	
		8	-0.09	0.00		-0.03		0.00	
		9	0.38	0.00		0.00		0.00	
		7	0.02	0.00		0.01		0.00	
		8	0.18	0.00		0.03		0.00	
		9	0.31	0.00		-0.01		0.00	
He et al., (2024)	1	10	0.33	0.01		0.11		0.01	TRUE
Hidalgo et al., (2021)	1	11	0.32	0.02		0.15		0.01	TRUE
		12	-0.11	0.00		-0.02		0.01	TRUE
Hristova et al., (2015)	1	13	0.48	0.03		0.16		0.01	TRUE
		14	0.45	0.02		0.15		0.01	TRUE

Article	Study	Ps ID	g	Cooks		DFBETAS		Hat	
				Value	Flag	Value	Flag	Value	Flag
Komatsu et al., (2017)	1	15	-0.20	0.00		-0.03		0.01	
		16	-0.24	0.00		-0.04		0.01	
Komatsu et al., (2021)	1	17	0.22	0.01		0.10		0.01	TRUE
	2	18	0.20	0.01		0.09		0.01	TRUE
	3	19	-0.04	0.00		0.00		0.01	TRUE
Laakasuo et al., (2021)	1	20	0.01	0.00		-0.03		0.00	
		20	0.44	0.01		0.08		0.01	
		21	-0.53	0.00		-0.03		0.00	
		21	-0.45	0.00		-0.02		0.01	
	2	22	-0.06	0.00		0.04		0.01	
		22	-0.45	0.01		-0.07		0.01	
		23	-0.04	0.00		0.02		0.01	
		23	-0.19	0.00		-0.03		0.01	
Laakasuo et al., (2023a)	1	24	-0.28	0.00		-0.01		0.01	
		24	-0.30	0.00		-0.01		0.01	
		25	0.06	0.00		-0.01		0.01	
		25	0.15	0.00		0.02		0.01	
Laakasuo et al., (2023b)	2	26	-0.15	0.00		0.00		0.01	
		27	0.28	0.00		0.03		0.01	
		26	-0.18	0.00		-0.01		0.01	
		27	0.15	0.00		0.00		0.01	
	3	28	-0.42	0.00		0.00		0.01	
		29	0.06	0.00		0.02		0.01	
		28	-0.47	0.00		-0.01		0.01	
		29	-0.04	0.00		-0.02		0.01	
	4	30	-0.45	0.00		-0.03		0.01	
		31	1.33	0.02		0.16		0.00	

Article	Study	Ps ID	g	Cooks		DFBETAS		Hat	
				Value	Flag	Value	Flag	Value	Flag
Malle et al., (2015)	5	32	-0.31	0.00		-0.02		0.01	
		30	-0.31	0.00		0.00		0.01	
		31	0.52	0.00		-0.03		0.01	
		32	-0.20	0.00		0.00		0.01	
		33	-0.10	0.00		0.05		0.01	
	1	34	0.29	0.00		0.01		0.01	
		33	-0.45	0.00		-0.06		0.01	
		34	0.30	0.00		0.01		0.01	
		35	0.17	0.00		0.05		0.01	TRUE
		36	0.49	0.01		0.12		0.01	
		2	37	1.04	0.05	0.23		0.01	
		38	-0.53	0.01		-0.09		0.01	
Malle et al., (2016)	2	39	0.18	0.01		0.09		0.01	TRUE
Malle et al., (2019)	1	40	-0.35	0.01		-0.08		0.01	TRUE
Malle et al., (2024)	2	41	0.19	0.01		0.08		0.01	TRUE
		42	0.13	0.00		0.05		0.01	TRUE
		43	-0.18	0.00		-0.02		0.01	
		44	-0.07	0.00		0.00		0.01	TRUE
		45	-0.21	0.00		-0.05		0.01	TRUE
		46	0.02	0.00		0.02		0.01	TRUE
		47	0.80	0.00		0.07		0.00	
		48	0.14	0.00		0.06		0.01	TRUE
		49	-0.17	0.00		-0.04		0.01	TRUE
		50	0.04	0.00		0.03		0.01	TRUE
		51	0.05	0.00		0.03		0.01	TRUE
		52	-0.16	0.00		-0.03		0.01	TRUE
		53	0.07	0.00		0.04		0.01	TRUE

Article	Study	Ps ID	g	Cooks		DFBETAS		Hat	
				Value	Flag	Value	Flag	Value	Flag
Maninger et al., (2022)	4	54	0.03	0.00		0.03		0.01	TRUE
		55	-0.03	0.00		0.01		0.01	TRUE
	2	56	-0.48	0.00		-0.02		0.00	
		56	-0.05	0.00		0.00		0.00	
		56	0.02	0.00		0.05		0.01	
		56	-0.25	0.00		-0.02		0.00	
Mayer et al., (2023)	1	57	-0.67	0.04		-0.20		0.01	TRUE
	2	58	-0.47	0.02		-0.15		0.01	TRUE
Meder et al., (2019)	1	59	0.29	0.02		0.14		0.01	TRUE
Shank et al., (2019)	1	60	0.25	0.01		0.09		0.01	TRUE
Soares et al., (2023)	1	61	0.11	0.00		0.02		0.00	
		61	0.11	0.00		0.02		0.00	
		62	0.01	0.00		0.00		0.00	
		62	0.01	0.00		0.00		0.00	
		61	-0.15	0.00		-0.02		0.00	
		61	-0.15	0.00		-0.02		0.00	
		62	-0.04	0.00		0.00		0.00	
		62	-0.04	0.00		0.00		0.00	
		63	-0.28	0.00		-0.01		0.00	
		63	-0.19	0.00		0.00		0.00	
Sundvall et al., (2023)	1	63	-0.16	0.00		0.01		0.00	
		64	-0.11	0.00		0.00		0.00	
		64	-0.11	0.00		0.00		0.00	
	2	64	-0.11	0.00		0.00		0.00	
		64	-0.11	0.00		0.00		0.00	
		64	-0.11	0.00		0.00		0.00	
		64	-0.11	0.00		0.00		0.00	
		64	-0.11	0.00		0.00		0.00	

Article	Study	Ps ID	g	Cooks		DFBETAS		Hat	
				Value	Flag	Value	Flag	Value	Flag
		64	-0.11	0.00		0.00		0.00	
		64	-0.11	0.00		0.00		0.00	
		64	-0.04	0.00		0.00		0.00	
		64	-0.27	0.00		0.00		0.00	
		64	-0.04	0.00		0.00		0.00	
		64	-0.27	0.00		0.00		0.00	
		64	-0.04	0.00		0.00		0.00	
		64	-0.27	0.00		0.00		0.00	
		64	-0.04	0.00		0.00		0.00	
		64	-0.27	0.00		0.00		0.00	
		64	-0.16	0.00		0.00		0.00	
		64	-0.27	0.00		0.00		0.00	
		64	-0.16	0.00		0.00		0.00	
		64	-0.27	0.00		0.00		0.00	
		64	-0.16	0.00		0.00		0.00	
		64	-0.27	0.00		0.00		0.00	
		64	-0.16	0.00		0.00		0.00	
		64	-0.27	0.00		0.00		0.00	
		64	-0.16	0.00		0.00		0.00	
		64	-0.27	0.00		0.00		0.00	
		65	-0.53	0.00		-0.01		0.00	
		65	-0.51	0.00		-0.01		0.00	
		65	-0.53	0.00		-0.01		0.00	
		65	-0.51	0.00		-0.01		0.00	
		65	-0.53	0.00		-0.01		0.00	
		65	-0.51	0.00		-0.01		0.00	
		65	-0.53	0.00		-0.01		0.00	
		65	-0.51	0.00		-0.01		0.00	
		65	0.15	0.00		0.01		0.00	

Article	Study	Ps ID	g	Cooks		DFBETAS		Hat	
				Value	Flag	Value	Flag	Value	Flag
		65	0.07	0.00		0.00		0.00	
		65	0.15	0.00		0.01		0.00	
		65	0.07	0.00		0.00		0.00	
		65	0.15	0.00		0.01		0.00	
		65	0.07	0.00		0.00		0.00	
		65	0.15	0.00		0.01		0.00	
		65	0.07	0.00		0.00		0.00	
		65	-0.08	0.00		0.00		0.00	
		65	-0.18	0.00		0.00		0.00	
		65	-0.08	0.00		0.00		0.00	
		65	-0.18	0.00		0.00		0.00	
		65	-0.08	0.00		0.00		0.00	
		65	-0.18	0.00		0.00		0.00	
		65	-0.08	0.00		0.00		0.00	
		65	-0.18	0.00		0.00		0.00	
		66	0.01	0.00		0.00		0.00	
		66	0.02	0.00		0.00		0.00	
		66	0.01	0.00		0.00		0.00	
		66	0.02	0.00		0.00		0.00	
		66	0.01	0.00		0.00		0.00	
		66	0.02	0.00		0.00		0.00	
		66	0.01	0.00		0.00		0.00	
		66	0.02	0.00		0.00		0.00	
		66	-0.02	0.00		0.00		0.00	
		66	0.03	0.00		0.00		0.00	
		66	-0.02	0.00		0.00		0.00	
		66	0.03	0.00		0.00		0.00	

Article	Study	Ps ID	g	Cooks		DFBETAS		Hat	
				Value	Flag	Value	Flag	Value	Flag
4		66	-0.02	0.00		0.00		0.00	
		66	0.03	0.00		0.00		0.00	
		66	-0.02	0.00		0.00		0.00	
		66	0.03	0.00		0.00		0.00	
		66	0.04	0.00		0.00		0.00	
		66	-0.13	0.00		0.00		0.00	
		66	0.04	0.00		0.00		0.00	
		66	-0.13	0.00		0.00		0.00	
		66	0.04	0.00		0.00		0.00	
		66	-0.13	0.00		0.00		0.00	
		66	0.04	0.00		0.00		0.00	
		66	-0.13	0.00		0.00		0.00	
		67	-0.24	0.00		0.00		0.00	
		67	-0.24	0.00		0.00		0.00	
		67	-0.24	0.00		0.00		0.00	
		67	-0.24	0.00		0.00		0.00	
		67	-0.24	0.00		0.00		0.00	
		67	-0.24	0.00		0.00		0.00	
		67	-0.24	0.00		0.00		0.00	
		67	-0.24	0.00		0.00		0.00	
		67	-0.24	0.00		0.00		0.00	
		67	-0.24	0.00		0.00		0.00	
		67	-0.24	0.00		0.00		0.00	
		67	-0.24	0.00		0.00		0.00	
		67	-0.19	0.00		0.00		0.00	
		67	-0.19	0.00		0.00		0.00	
		67	-0.19	0.00		0.00		0.00	

[illegible]

[illegible]

[illegible]

Article	Study	Ps ID	g	Cooks		DFBETAS		Hat	
				Value	Flag	Value	Flag	Value	Flag
Wilson et al., (2022)	1	69	-0.22	0.00		0.00		0.00	
		69	-0.22	0.00		0.00		0.00	
		69	-0.22	0.00		0.00		0.00	
		69	-0.22	0.00		0.00		0.00	
		69	-0.22	0.00		0.00		0.00	
		69	-0.22	0.00		0.00		0.00	
		69	-0.22	0.00		0.00		0.00	
		69	-0.22	0.00		0.00		0.00	
		69	-0.22	0.00		0.00		0.00	
		69	-0.22	0.00		0.00		0.00	
		69	-0.22	0.00		0.00		0.00	
		69	-0.22	0.00		0.00		0.00	
		70	0.20	0.00		0.01		0.00	
		70	0.10	0.00		0.00		0.00	
70	0.11	0.00		0.00		0.00			
Young et al., (2019)	2	71	0.09	0.00		0.05		0.01	TRUE
	1	72	-0.36	0.01		-0.09		0.01	TRUE
		73	-0.23	0.00		-0.05		0.01	TRUE
	2	74	-0.22	0.00		-0.05		0.01	TRUE
		75	-0.16	0.00		-0.03		0.01	TRUE
	3	76	-0.13	0.00		-0.02		0.01	TRUE
		77	-0.07	0.00		0.00		0.01	TRUE
Zhang et al., (2022)	1	78	0.20	0.00		0.07		0.01	TRUE
Zhang et al., (2023)	1	79	-2.11	0.00		-0.04		0.00	
		79	-2.15	0.00		-0.04		0.00	
		79	-0.78	0.00		0.02		0.00	
		79	-0.51	0.00		0.04		0.00	

Article	Study	Ps ID	g	Cooks		DFBETAS		Hat	
				Value	Flag	Value	Flag	Value	Flag
	2	79	-1.65	0.00		-0.03		0.00	
		79	-0.57	0.00		0.03		0.00	
		80	0.72	0.00		0.01		0.00	
		80	0.16	0.00		-0.04		0.00	
		80	-1.60	0.03		-0.16		0.00	
		80	2.02	0.01		0.09		0.00	
	3	80	1.69	0.01		0.08		0.00	
		80	1.10	0.00		0.05		0.00	
		81	-0.88	0.00		-0.02		0.00	
		81	-2.07	0.00		-0.05		0.00	
		81	0.95	0.00		0.07		0.00	
		81	-0.99	0.00		-0.02		0.00	
		81	-4.16	0.00		-0.06		0.00	
		81	0.77	0.00		0.06		0.00	
		81	1.62	0.01		0.08		0.00	
		81	-2.00	0.00		-0.05		0.00	
		81	-1.23	0.00		-0.03		0.00	

1.3 I^2

Figure S3

Plot Of I^2 Showing How Much Variation Is Explained By Within-PIDs And Between-PIDs Heterogeneity In The Full Data.

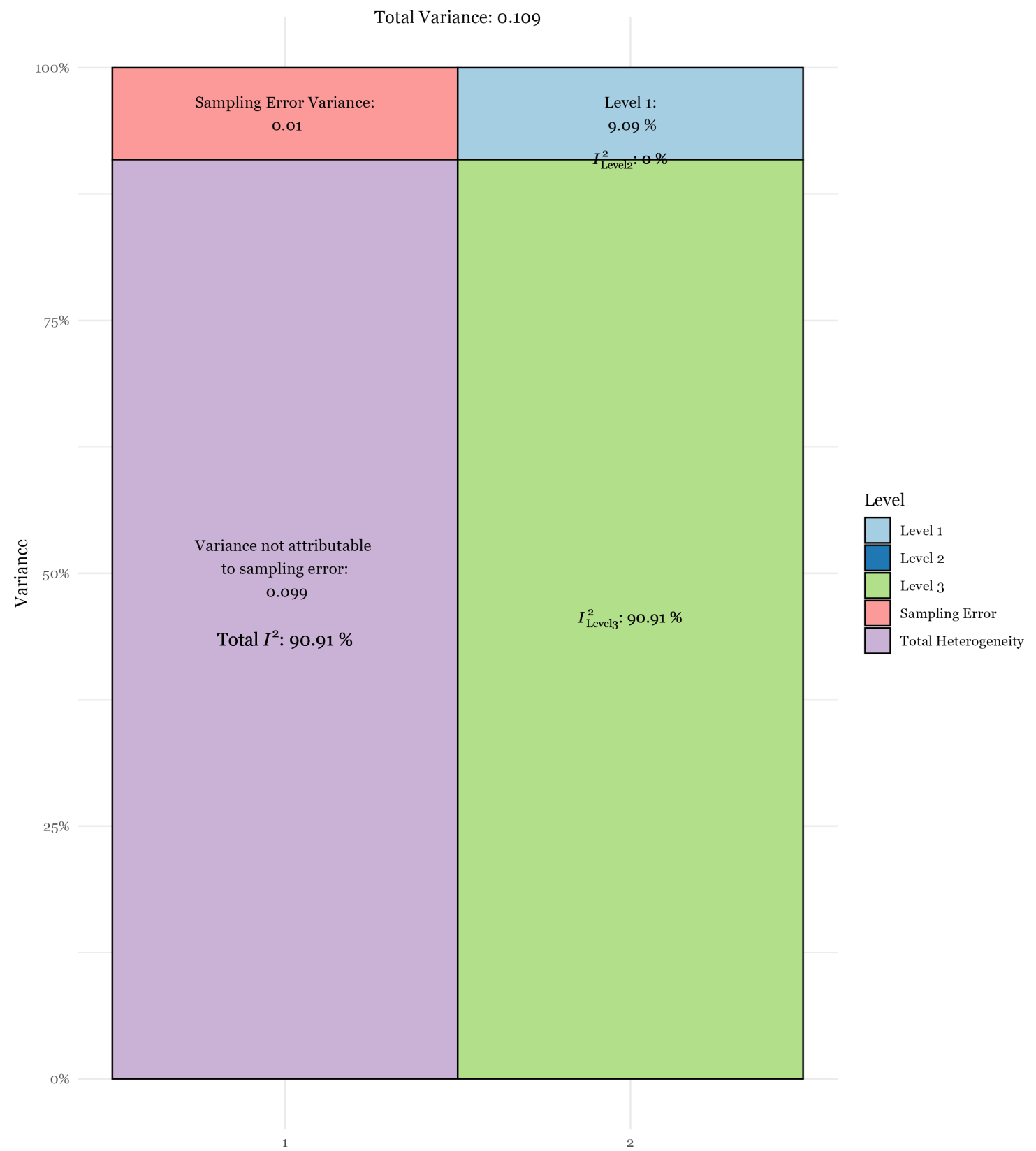
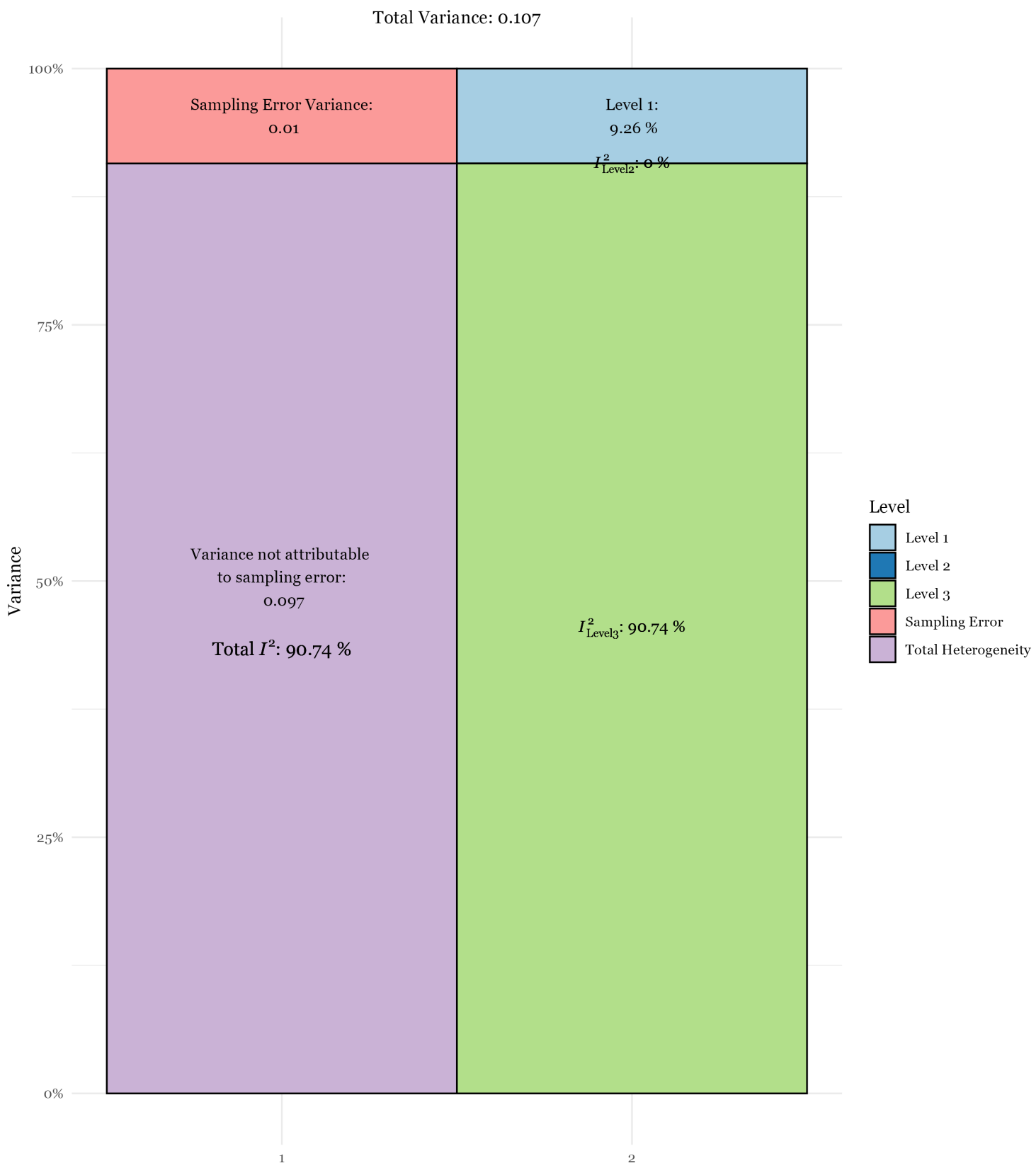


Figure S4

Plot Of I^2 Showing How Much Variation Is Explained By Within-PIDs And Between-PIDs Heterogeneity In The Trimmed Data.



1.4 Iterative Random-Effects Model Fit Analyses

Table S2

Fit Statistics for All Random-Effects Nesting Structure Combinations (Full and Trimmed Datasets)

Random Effects		Model Fit Index			g	sigma ²		
Levels	Nesting	AIC	BIC	logLik		1	2	3
Full Data								
Four	S / Ps / ES	214.42	230.06	-102.59	-0.07	0.000	0.099	0.000
	A / S / Ps	211.98	227.61	-101.07	-0.06	0.031	0.000	0.073
Three	S / Ps	212.42	224.15	-102.59	-0.07	0.000	0.099	
	A / ES	343.16	354.88	-167.57	-0.08	0.053	0.073	
	A / P	209.98	221.70	-101.07	-0.06	0.031	0.073	
	Ps / ES	212.42	224.15	-102.59	-0.07	0.099	0.000	
	S / ES	270.12	281.84	-131.26	-0.11	0.090	0.029	
Two	ES	431.47	439.29	-213.75	-0.23	0.135		
	Ps	210.42	218.24	-102.59	-0.07	0.099		
	S	289.60	297.42	-142.04	-0.10	0.093		
One	None	1,526.79	1,530.70	-763.79	-0.25	0.000		
Trimmed Data								
Four	S / Ps / ES	85.95	101.53	-38.34	-0.08	0	0.097	0
	A / S / Ps	83.69	99.26	-36.88	-0.08	0.031	0	0.072
Three	S / Ps	83.95	95.64	-38.34	-0.08	0	0.097	
	A / ES	257.15	268.83	-124.57	-0.09	0.051	0.054	
	A / P	81.69	93.37	-36.88	-0.08	0.031	0.072	
	Ps / ES	83.95	95.64	-38.34	-0.08	0.097	0	
	S / ES	155.02	166.7	-73.74	-0.12	0.09	0	
Two	ES	359.03	366.82	-177.61	-0.23	0.112		
	Ps	81.95	89.74	-38.34	-0.08	0.097		
	S	153.02	160.81	-73.74	-0.12	0.09		
One	None	1,273.03	1,276.93	-636.91	-0.25	0		

1.5 Intent x Decision Type (Full Output for Trimmed Data)

Supplemental Result S1

Additional Moderator Outputs For Intent And Decision Type Models

The NHST results for all levels of the intent and decision type moderators in the multiple moderator model:

Intent:

Means-to-end: $n_{\text{AI}} = 1777$, $n_{\text{Human}} = 1534$, $k_{\text{PIDs}} = 11$, $k_{\text{comp}} = 21$, $g = -0.07$, 95% CI = $[-0.25, .10]$, $p = .396$

Side-effect: $n_{\text{AI}} = 64673$, $n_{\text{Human}} = 64393$, $k_{\text{PIDs}} = 43$, $k_{\text{comp}} = 273$, $g = .13$, 95% CI = $[-.01, .26]$, $p = .032$

Decision Type:

Action: $n_{\text{AI}} = 63480$, $n_{\text{Human}} = 63554$, $k_{\text{PIDs}} = 38$, $k_{\text{comp}} = 267$, $g = .10$, 95% CI = $[-0.07, .27]$, $p = .267$

Inaction: $n_{\text{AI}} = 2970$, $n_{\text{Human}} = 2373$, $k_{\text{PIDs}} = 17$, $k_{\text{comp}} = 27$, $g = -0.10$, 95% CI = $[-0.27, .07]$, $p = .267$

The TOST results:

Intent

Means-to-end: $g = 0.07$, 90% CI $[-0.07, 0.22]$, bounds ± 0.16 , $p = 0.176$

Side-effect: $g = -0.13$, 90% CI $[-0.24, -0.03]$, bounds ± 0.16 , $p = 0.356$

Decision Type

Action: $g = 0.10$, 90% CI $[-0.05, 0.24]$, bounds ± 0.15 , $p = 0.286$

Inaction: $g = -0.10$, 90% CI $[-0.24, 0.05]$, bounds ± 0.23 , $p = 0.069$

1.6 Responsibility Attributions Moderator, Controlling for Wording

Supplemental Result S2

Additional Moderator Model Controlling for Responsible Categories.

Moderation effect: $F_{4,298}=0.07$, $p= 0.991$

Heterogeneity test: $Q(df=298)=927.76$, $p\leq .001$

For the moderation effect of responsibility attributions scores alone (whilst controlling for the wording/phrasing categories):

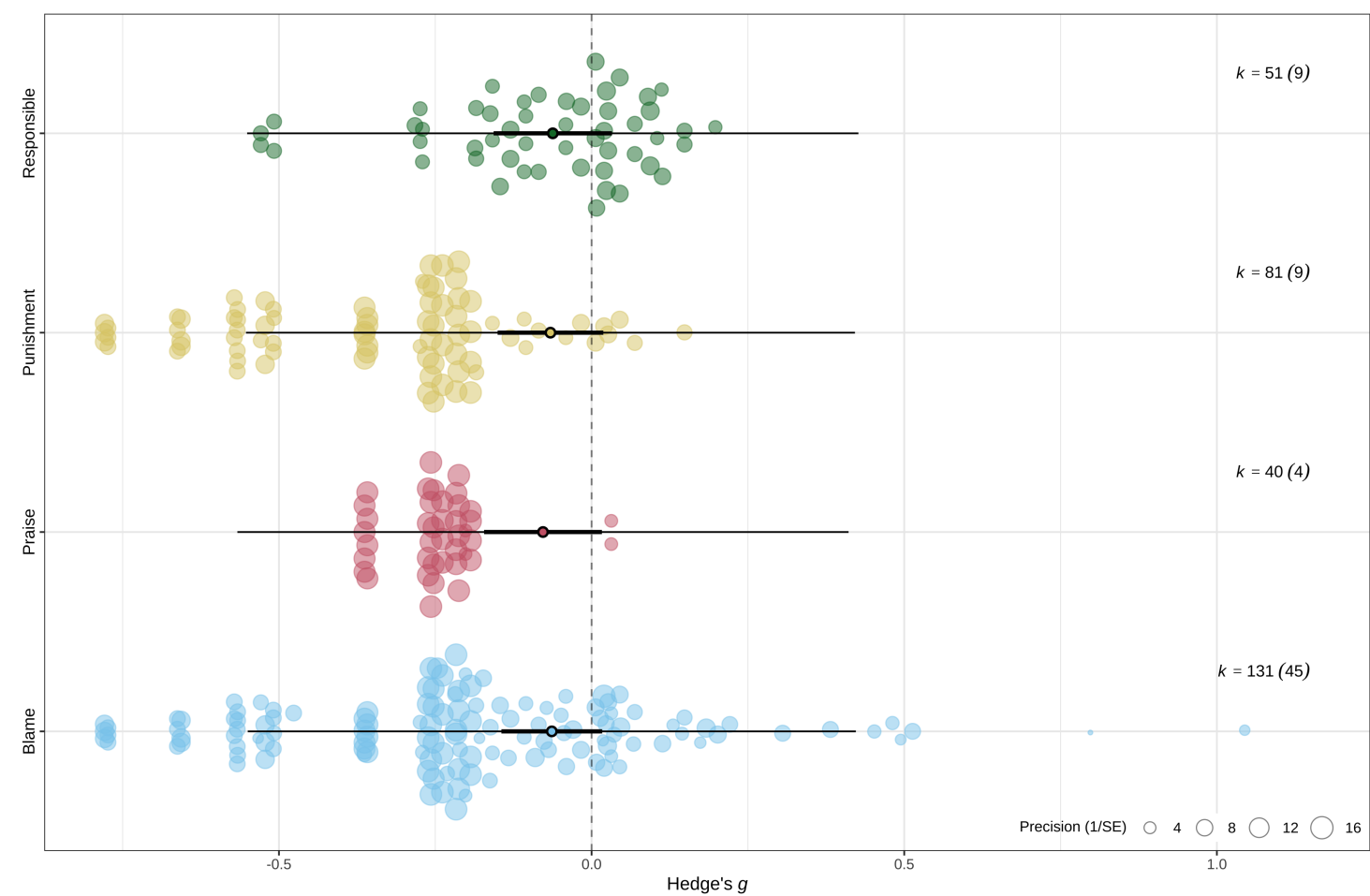
NHST: $k_{\text{PIDs}} = 49$, $k_{\text{comp}} = 303$, $g = -0.01$, 95% CI = $[-0.08, .05]$, $p = .654$

TOST: $g = -0.00$, 90% CI $[-0.03, 0.02]$, bounds ± 0.04 , $p = 0.003$

The orchard plot in Figure S5 visualises relatively consistent distributions across the four responsibility attribution wordings.

Figure S5

Orchard Plot of Responsibility Attributions broken down by Wording Categories



1.7 Moderator Breakdown by Effect Size Comparisons

Table S3 details the moderator levels and/or values for each comparison (effect size ID).

Table S3

Levels of Moderators Each Study Contributes to.

First Author, Year	Study	PsID	EFIDs	AI <i>N</i>	Human <i>N</i>	DV* (syn- onym)	<i>g</i>	<i>Var</i>	Intent	Decision Type	Moral Do- main	Agent Intel- li- gence	2- Category AI	3- Category AI	PMA	PMC	Responsibility RQ	
Banks et al., (2021)	Study 1	1	1, 2	194	208	G	0.09	0.01			H	NI	R	Hum.	-0.53, 0.22		-0.46	59.50
		1	3, 4	194	208	G	0.02	0.01			NH	NI	R	Hum.	-0.53, 0.22		-0.46	59.50
	Study 2	2	5-8	58	34	G	0.03	0.05			H	NI	R	Hum.	-0.29, 0.37	-1.20	0.06, -0.44	59.50
		2	9-12	58	34	G	-0.20	0.05			NH	NI	R	Hum.	-0.29, 0.37	-1.20	0.06, -0.44	59.50
Bigman et al., (2018)	Study 3	3	13-18	108	118	M	-0.57	0.02	S-E	A	H	I	AI	AI	-0.66, -0.75, -0.69	-3.45	-0.66, -1.02	59.50
		3	19-24	108	118	R	-0.57	0.02	S-E	A	H	I	AI	AI	-0.66, -0.75, -0.69	-3.45	-0.66, -1.02	59.50
		3	25-30	108	118	P	-0.51	0.02	S-E	A	H	I	AI	AI	-0.66, -0.75, -0.69	-3.45	-0.66, -1.02	59.50
	Study 4	4	31-36	119	114	M	-0.66	0.02	S-E	A	H	I	AI	AI	-1.05, -1.10, 1.04	-2.07	-0.45, -0.73	61.00
		4	37-42	119	114	R	-0.57	0.02	S-E	A	H	I	AI	AI	-1.05, -1.10, 1.04	-2.07	-0.45, -0.73	61.00

	Study 5	4	43-48	119	114	P	-0.77	0.02	S-E	A	H	I	AI	AI	-1.05, -1.10, 1.04	-2.07	-0.45, -0.73	61.00
		5	49-54	224	231	M	-0.66	0.01		A	H	I	AI	AI	-1.14, -1.58, -1.33	-2.79	-0.33, -0.54	59.50
		5	55-60	224	231	R	-0.52	0.01		A	H	I	AI	AI	-1.14, -1.58, -1.33	-2.79	-0.33, -0.54	59.50
		5	61-66	224	231	P	-0.78	0.01		A	H	I	AI	AI	-1.14, -1.58, -1.33	-2.79	-0.33, -0.54	59.50
Cassioli et al., (2023)	Study 1	6	67	34	34	M	-1.08	0.14			H	NI	AI	AI				48.00
Chu et al., (2023)	Study 2	7	68	121	127	P	-0.08	0.02	S-E	A	H	NI	R	Mech.			0.05	61.00
		8	69	218	206	P	0.05	0.01	S-E	I	H	NI	R	Mech.			0.08	61.00
		9	70	122	110	P	0.51	0.02	M-E	A	H	NI	R	Mech.			-0.45	61.00
		7	71	121	127	W	0.01	0.02	S-E	A	H	NI	R	Mech.			0.05	61.00
		8	72	218	206	W	-0.09	0.01	S-E	I	H	NI	R	Mech.			0.08	61.00
		9	73	122	110	W	0.38	0.02	M-E	A	H	NI	R	Mech.			-0.45	61.00
		7	74	121	127	Ac	0.02	0.02	S-E	A	H	NI	R	Mech.			0.05	61.00
		8	75	218	206	Ac	0.18	0.01	S-E	I	H	NI	R	Mech.			0.08	61.00
		9	76	122	110	Ac	0.31	0.02	M-E	A	H	NI	R	Mech.			-0.45	61.00
He et al., (2024)	Study 1	10	77	40	40	M	0.33	0.05			NH	NI	AI	AI				56.00
Hristova et al., (2015)	Study 1	11	80	53	53	P	0.48	0.04		A	H	I	AI	AI			-0.06	37.00
		12	81	53	53	P	0.45	0.04		A	H	I	R	Hum.			-0.50	37.00
Komatsu et al., (2017)	Study 1	13	82	35	35	W	-0.20	0.11	S-E	A	H	I	R	Mech.				31.00
		14	83	34	37	W	-0.24	0.10	S-E	I	H	I	R	Mech.				31.00

Komatsu et al., (2021)	Study 1	15	84	126	142	P	0.22	0.02	S-E	A	H	I	R	Mech.		0.05	-0.13	53.00
	Study 2	16	85	194	191	P	0.20	0.01	S-E	A	H	I	R	Mech.		-0.06	-0.16	52.00
	Study 3	17	86	155	94	P	-0.04	0.02	S-E	A	H	I	R	Mech.		-0.11	-0.12	52.00
Laakasuo et al., (2021)	Study 1	18	87	21	26	M	0.01	0.09	M-E	A	H	NI	R	Mech.	-0.51	-1.78		47.00
		18	88	38	26	M	0.44	0.07	M-E	A	H	NI	R	Hum.	-0.06	-1.94		47.00
		19	89	19	21	M	-0.53	0.10	S-E	I	H	NI	R	Mech.	-0.04	-1.79		47.00
		19	90	35	21	M	-0.45	0.08	S-E	I	H	NI	R	Hum.	-0.41	-2.44		47.00
Laakasuo et al., (2023)	Study 2	20	91	49	50	M	-0.06	0.04	M-E	A	H	NI	R	Mech.	-0.40	-1.74		50.00
		20	92	100	50	M	-0.45	0.03	M-E	A	H	NI	R	Hum.	-0.43	-1.15		50.00
		21	93	50	52	M	-0.04	0.04	S-E	I	H	NI	R	Mech.	-0.01	-1.58		50.00
		21	94	97	52	M	-0.19	0.03	S-E	I	H	NI	R	Hum.	-0.26	-1.39		50.00
	Study 1	22	95	182	62	M	-0.28	0.02	M-E	A	H	NI	R	Mech.	-0.81	-3.64		66.00
		22	96	66	62	M	-0.30	0.03	M-E	A	H	NI	R	Hum.	-1.35	-0.54		66.00
		23	97	189	62	M	0.06	0.02	S-E	I	H	NI	R	Mech.	-0.31	-3.03		66.00
		23	98	64	62	M	0.15	0.03	S-E	I	H	NI	R	Hum.	-0.66	-0.62		66.00
	Study 2	24	99	32	37	Ap	-0.15	0.06		A	H	I	R	Mech.				49.00
		25	100	29	37	Ap	0.28	0.06		I	NH	I	R	Mech.				49.00
		24	101	32	37	R	-0.18	0.06		A	H	I	R	Mech.				49.00
		25	102	29	37	R	0.15	0.06		I	NH	I	R	Mech.				49.00
	Study 3	26	103	233	246	Ap	-0.42	0.01		A	H	I	R	Mech.				70.00
		27	104	243	259	Ap	0.06	0.01		I	NH	I	R	Mech.				70.00
		26	105	233	246	R	-0.47	0.01		A	H	I	R	Mech.				70.00
		27	106	243	259	R	-0.04	0.01		I	NH	I	R	Mech.				70.00
	Study 4	28	107	35	41	Ap	-0.45	0.05	S-E	A	H	I	R	Mech.				67.00
		29	108	29	28	Ap	1.33	0.09	S-E	A	H	I	R	Mech.				67.00
		30	109	30	42	Ap	-0.31	0.06	Na	A	H	I	R	Mech.				67.00
		28	110	35	41	R	-0.31	0.05	S-E	A	H	I	R	Mech.				67.00
		29	111	29	28	R	0.52	0.07	S-E	A	H	I	R	Mech.				67.00
		30	112	30	42	R	-0.20	0.06	Na	A	H	I	R	Mech.				67.00

Malle et al., (2015)	Study 5	31	113	126	123	Ap	-0.10	0.02		A	H	I	R	Mech.			54.00
		32	114	124	127	Ap	0.29	0.02		I	NH	I	R	Mech.			54.00
		31	115	126	123	R	-0.45	0.02		A	H	I	R	Mech.			54.00
		32	116	124	127	R	0.30	0.02		I	NH	I	R	Mech.			54.00
	Study 1	33	117	39	40	P	0.17	0.07	S-E	A	H	I	R	Mech.	0.05	-0.11	46.50
Malle et al., (2019)	Study 2	34	118	40	38	P	0.49	0.09	S-E	I	H	I	R	Mech.	-0.20	0.29	46.50
		35	119	40	39	W	1.04	0.10	S-E	A	H	I	R	Mech.	0.04	-0.54	46.50
	36	120	41	40	W	-0.53	0.09	S-E	I	H	I	R	Mech.	-0.17	0.57	46.50	
	Study 1	37	122	270	89	W	-0.35	0.03	M-E	A	H	I	AI	AI			52.00
	Malle et al., (2024)	Study 2.1	38	123	269	90	W	0.19	0.02	S-E	I	H	I	AI	AI		
39			124	51	49	W	0.13	0.05		A	H	I	R	Mech.		-0.08	52.00
Study 2.2		40	125	48	49	W	-0.18	0.09		I	H	I	R	Mech.		-0.17	52.00
		41	126	68	71	W	-0.07	0.04	M-E	A	H	I	R	Mech.		-0.04	51.00
Study 2.3		42	127	140	139	W	-0.21	0.02	S-E	I	H	I	R	Mech.		0.14	51.00
		43	128	71	69	W	0.02	0.08	S-E	A	H	I	R	Mech.		0.02	51.00
		44	129	69	68	W	0.80	0.39		A	H	I	R	Mech.		-0.07	51.00
Study 2.4		45	130	80	77	W	0.14	0.04	M-E	A	H	I	R	Mech.		0.00	54.00
Study 4.1		46	131	239	239	W	-0.17	0.01	S-E	I	H	I	R	Mech.		0.18	54.00
		47	132	160	152	W	0.04	0.02	S-E	A	H	I	R	Mech.		-0.09	54.00
		48	133	81	80	W	0.05	0.03	M-E	A	H	I	R	Mech.		-0.17	54.00
Study 4.3	49	134	188	190	W	-0.16	0.02	S-E	I	H	I	R	Mech.		0.23	54.00	
	50	135	107	108	W	0.07	0.03	S-E	A	H	I	R	Mech.		-0.39	54.00	
	51	136	453	330	W	0.03	0.01			H	I	R	Mech.		-0.01	54.00	

Maninger et al., (2022)	Study 2	53	138	91	145	W	-0.48	0.02	M-E	A	H	NI	AI	AI		1.20	44.50
		53	139	44	145	W	-0.05	0.03	M-E	A	H	NI	R	Mech.		1.83	44.50
		53	140	407	725	W	0.02	0.00		A	NH	NI	AI	AI		1.09	44.50
		53	141	226	725	W	-0.25	0.01		A	NH	NI	R	Mech.		1.01	44.50
Mayer et al., (2023)	Study 1	54	142	173	187	J	-0.67	0.01	S-E	A	H	NI	AI	AI			52.50
	Study 2	55	143	507	248	J	-0.47	0.01	S-E	A	H	NI	AI	AI			50.50
Shank et al., (2019)	Study 1	56	145	58	58		0.25	0.03		A	NH	NI	AI	AI			44.00
Soares et al., (2023)	Study 1	57	146, 147	134	136	Ac	0.11	0.01		A	NH	NI	R	Mech.	-0.09	-0.39, -0.02	81.00
		58	148, 149	130	122	Ac	0.01	0.02		I	NH	NI	R	Mech.	-0.10	-0.30, -0.02	81.00
		57	150, 151	134	137	R	-0.15	0.01		A	NH	NI	R	Mech.	-0.09	-0.39, -0.02	81.00
		58	152, 153	130	122	R	-0.04	0.02		I	NH	NI	R	Mech.	-0.10	-0.30, -0.02	81.00
Sundvall et al., (2023)	Study 1	59	154	115	101	Ac	-0.28	0.02	S-E	A	H	NI	R	Mech.		-0.19	56.00
		59	155	115	101	W	-0.19	0.02	S-E	A	H	NI	R	Mech.		-0.19	56.00
		59	156	115	101	U	-0.16	0.02	S-E	A	H	NI	R	Mech.		-0.19	56.00
	Study 2.1	60	157, 159, 161, 163	59	65	Ac	-0.11	0.03	S-E	A	H	NI	R	Mech.	0.71	-0.87, -1.60, -0.23, -0.44	63.00
		60	158, 160, 162, 164	62	65	Ac	-0.11	0.03	S-E	A	H	NI	R	Hum.	0.21	-0.15, -0.57, -0.20, 0.14	63.00

Study 2.2	60	165, 167, 169, 171	59	65	W	-0.04	0.03	S-E	A	H	NI	R	Mech.	0.71	-0.87, -1.60, -0.23, -0.44	63.00
	60	166, 168, 170, 172	62	65	W	-0.27	0.03	S-E	A	H	NI	R	Hum.	0.21	-0.15, -0.57, -0.20, 0.14	63.00
	60	173, 175, 177, 179	59	65	U	-0.16	0.03	S-E	A	H	NI	R	Mech.	0.71	-0.87, -1.60, -0.23, -0.44	63.00
	60	174, 176, 178, 180	62	65	U	-0.27	0.03	S-E	A	H	NI	R	Hum.	0.21	-0.15, -0.57, -0.20, 0.14	63.00
	61	181, 183, 185, 187	89	93	Ac	-0.53	0.02	S-E	A	H	NI	R	Mech.	0.16	-0.28, -0.55, 0.01, -0.70	61.00
	61	182, 184, 186, 188	84	93	Ac	-0.51	0.02	S-E	A	H	NI	R	Hum.	0.59	-0.77, -0.95, 0.02, -1.01	61.00
	61	189, 191, 193, 195	89	93	W	0.15	0.02	S-E	A	H	NI	R	Mech.	0.16	-0.28, -0.55, 0.01, -0.70	61.00
	61	190, 192, 194, 196	84	93	W	0.07	0.02	S-E	A	H	NI	R	Hum.	0.59	-0.77, -0.95, 0.02, -1.01	61.00

Study 2.3	61	197, 199, 201, 203	89	93	U	-0.08	0.02	S-E	A	H	NI	R	Mech.	0.16	-0.28, -0.55, 0.01, -0.70	61.00
	61	198, 200, 202, 204	84	93	U	-0.18	0.02	S-E	A	H	NI	R	Hum.	0.59	-0.77, -0.95, 0.02, -1.01	61.00
	62	205, 207, 209, 211	152	150	Ac	0.01	0.01	S-E	A	H	NI	R	Mech.	0.49	-0.23, -0.84, -0.11, -0.08	64.00
	62	206, 208, 210, 212	144	150	Ac	0.02	0.01	S-E	A	H	NI	R	Hum.	0.51	0.06, -0.54, -0.05, 0.12	64.00
	62	213, 215, 217, 219	152	150	W	-0.02	0.01	S-E	A	H	NI	R	Mech.	0.49	-0.23, -0.84, -0.11, -0.08	64.00
	62	214, 216, 218, 220	144	150	W	0.03	0.01	S-E	A	H	NI	R	Hum.	0.51	0.06, -0.54, -0.05, 0.12	64.00
	62	221, 223, 225, 227	152	150	U	0.04	0.01	S-E	A	H	NI	R	Mech.	0.49	-0.23, -0.84, -0.11, -0.08	64.00
	62	222, 224, 226, 228	144	150	U	-0.13	0.01	S-E	A	H	NI	R	Hum.	0.51	0.06, -0.54, -0.05, 0.12	64.00

Wilson et al., (2022)	Study 4.1	63	229-240	456	456	Ac	-0.24	0.00	S-E	A	H	NI	R	Hum.	-1.01, -0.99	0.14, 0.36, -0.80	86.50
		63	241-252	456	456	W	-0.19	0.00	S-E	A	H	NI	R	Hum.	-1.01, -0.99	0.14, 0.36, -0.80	86.50
		63	253-264	456	456	P	-0.21	0.00	S-E	A	H	NI	R	Hum.	-1.01, -0.99	0.14, 0.36, -0.80	86.50
	Study 4.2	64	265-276	403	408	Ac	-0.36	0.01	S-E	A	H	NI	R	Hum.	-0.54, -0.55	-0.12, 0.25, -0.65	59.50
		64	277-288	403	408	W	-0.25	0.00	S-E	A	H	NI	R	Hum.	-0.54, -0.55	-0.12, 0.25, -0.65	59.50
		64	289-300	403	408	P	-0.36	0.01	S-E	A	H	NI	R	Hum.	-0.54, -0.55	-0.12, 0.25, -0.65	59.50
	Study 5	65	301-312	454	461	Ac	-0.26	0.00	S-E	A	H	NI	R	Hum.	-0.70, -0.73	0.24, 0.23, -0.60	83.50
		65	313-324	454	461	W	-0.26	0.00	S-E	A	H	NI	R	Hum.	-0.70, -0.73	0.24, 0.23, -0.60	83.50
		65	325-336	454	461	P	-0.22	0.00	S-E	A	H	NI	R	Hum.	-0.70, -0.73	0.24, 0.23, -0.60	83.50
	Study 1	66	337	51	50	W	0.20	0.04			NH	NI	AI	AI		-0.63	54.50
		66	338	51	50	J	0.10	0.04			NH	NI	AI	AI		-0.63	54.50
		66	339	51	50	P	0.11	0.04			NH	NI	AI	AI		-0.63	54.50
	Study 2	67	340	159	154	W	0.09	0.01			NH	NI	AI	AI		-1.17	57.00

Young et al., (2019)	Study 1	68	341	101	101	P	-0.36	0.03	S-E	A	H	I	AI	AI	-1.31		0.16	42.00
		69	342	100	102	P	-0.23	0.03	S-E	I	H	I	AI	AI	-1.32		0.50	42.00
	Study 2	70	343	202	101	P	-0.22	0.02	S-E	A	H	I	AI	AI	-0.66		0.29	43.00
		71	344	200	100	P	-0.16	0.02	S-E	I	H	I	AI	AI	-0.24		0.09	43.00
	Study 3	72	345	199	99	P	-0.13	0.02	S-E	A	H	I	AI	AI	-0.85	-0.32	0.31	43.00
		73	346	202	101	P	-0.07	0.02	S-E	I	H	I	AI	AI	-0.71	-0.40	0.27	43.00
Zhang et al., (2022)	Study 1	74	347	312	312	W	0.20	0.05					R	Mech.				54.00
Zhang et al., (2023)	Study 1	75	348	49	49	M	-2.11	0.06	S-E	A	H	NI	AI	AI				45.00
		75	349	49	49	M	-2.15	0.06	S-E	I	H	NI	AI	AI				45.00
		75	350	49	49	P	-0.78	0.04	S-E	A	H	NI	AI	AI				45.00
		75	351	49	49	P	-0.51	0.04	S-E	I	H	NI	AI	AI				45.00
		75	352	49	49	W	-1.65	0.05	S-E	A	H	NI	AI	AI				45.00
		75	353	49	49	W	-0.57	0.04	S-E	I	H	NI	AI	AI				45.00
	Study 2	76	354	48	48	M	0.72	0.04	M-E	A	H	NI	AI	AI				45.00
		76	355	48	48	M	0.16	0.04	S-E	I	H	NI	AI	AI				45.00
		76	356	48	48	P	-1.60	0.06	M-E	A	H	NI	AI	AI				45.00
		76	357	48	48	P	2.02	0.06	S-E	I	H	NI	AI	AI				45.00
		76	358	48	48	W	1.69	0.06	M-E	A	H	NI	AI	AI				45.00
		76	359	48	48	W	1.10	0.05	S-E	I	H	NI	AI	AI				45.00
	Study 3	77	360	59	59	M	-0.88	0.04	S-E	A	H	NI	AI	AI				45.50
		77	361	59	59	M	-2.07	0.05	S-E	I	H	NI	AI	AI				45.50
		77	362	59	59	M	0.95	0.04	M-E	A	H	NI	AI	AI				45.50
		77	363	59	59	P	-0.99	0.04	S-E	A	H	NI	AI	AI				45.50
		77	365	59	59	P	0.77	0.04	M-E	A	H	NI	AI	AI				45.50
		77	366	59	59	W	1.62	0.05	S-E	A	H	NI	AI	AI				45.50
		77	367	59	59	W	-2.00	0.05	S-E	I	H	NI	AI	AI				45.50
		77	368	59	59	W	-1.23	0.04	M-E	A	H	NI	AI	AI				45.50

Note. text

2 Results for Full Data

2.1 Moderator Analyses

As with the analyses for the trimmed data reported in the manuscript, we report conventional NHST results alongside TOST results to provide converging evidence for either equivalence or non-equivalence. The TOST bounds were again defined using the MDES. For categorical moderators, MDES values were calculated separately for each level, reflecting the differing amounts of data available and therefore the differing statistical power associated with each level.

2.1.1 Moral Domain (Harm vs. Non-Harm Domains)

Moderation effect: $F_{1,365}=0.91, p= 0.341$
Heterogeneity test: $Q_{(df=365)}=2276.19, p\leq .001$

Table S4

Effects of Each Moral Domain Moderator Levels.

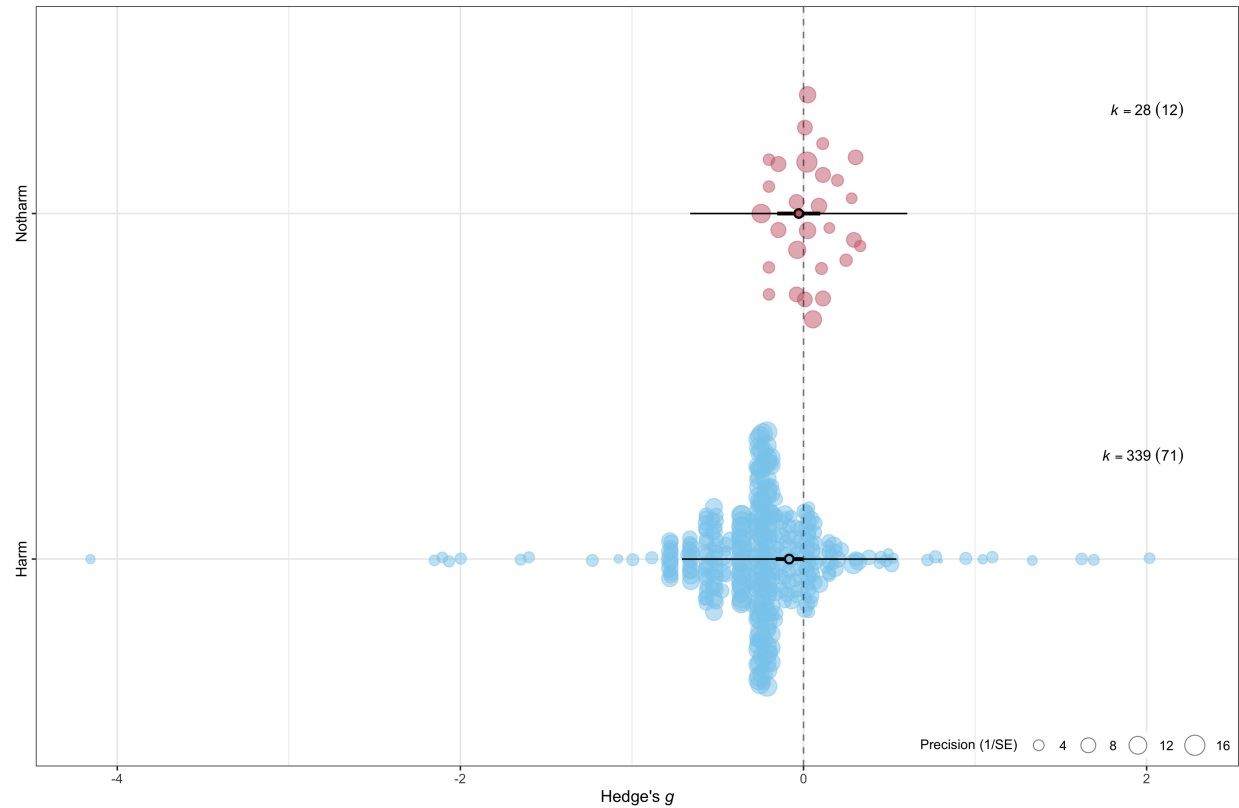
Term	nexp	ctrl	kcomp	kstudies	estimate	se	tval	df	pval	ci.lb	ci.ub
factor(harm)harm	74442	73819	339	71	-0.0834384	0.0394743	-2.1137396	365	0.0352154	-0.1610641	-0.0058128
factor(harm)noharm	3511	4284	28	12	-0.0278286	0.0630351	-0.4414773	365	0.6591289	-0.1517860	0.0961289

TOST results for each moderator level:

Harm: $g = -0.08, 90\% \text{ CI } [-0.15, -0.02], \text{ bounds } \pm 0.12, p = 0.189$
Not Harm: $g = -0.03, 90\% \text{ CI } [-0.13, 0.08], \text{ bounds } \pm 0.27, p = 0.000$

Figure S6

Orchard Plot of Moral Domain Moderator



2.1.2 Intent (Means-End vs. Side-Effect)

Moderation effect: $F_{1,297}=8.64, p= 0.004$

Heterogeneity test: $Q_{(df=297)}=1754.91, p\leq .001$

Table S5

Effects of Each Intent Moderator Levels.

Term	nexp	ctrl	kcomp	kstudies	estimate	se	tval	df	pval	ci.lb	ci.ub
factor(intent)meane	1777	1534	21	11	0.1360925	0.0982402	1.385304	297	0.1669990	-0.0572425	0.3294275
factor(intent)sidee	66110	65830	278	47	-0.1617149	0.0545102	-2.966693	297	0.0032548	-0.2689901	-0.0544398

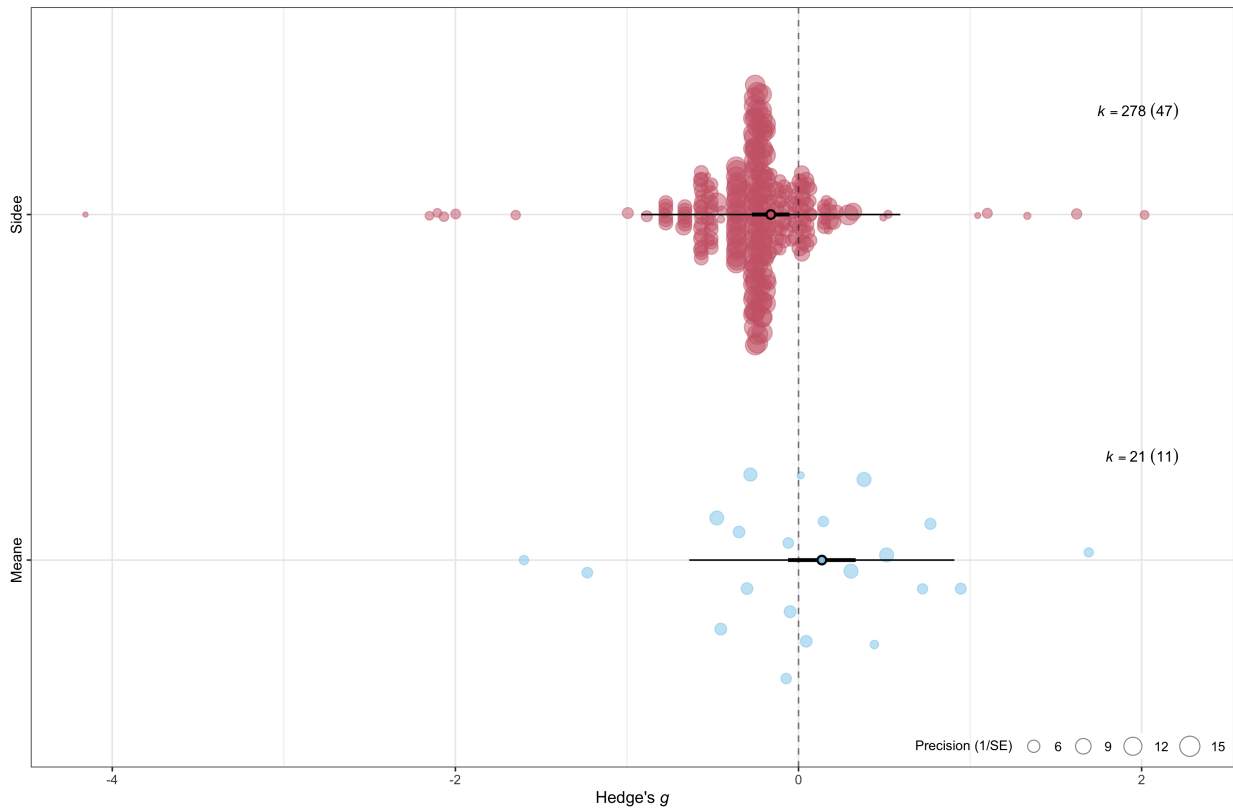
TOST results for each moderator level:

Means-to-end: $g = 0.14, 90\% \text{ CI } [-0.03, 0.30], \text{ bounds } \pm 0.33, p = 0.027$

Side-effect: $g = -0.16, 90\% \text{ CI } [-0.25, -0.07], \text{ bounds } \pm 0.16, p = 0.547$

Figure S7

Orchard Plot of Intent Moderator



2.1.3 Decision Type (Action vs. Inaction)

Moderation effect: $F_{1,344}=15.81, p< .001$

Heterogeneity test: $Q_{(df=344)}=2229.48, p\leq .001$

Table S6

Effects of Each Decision Type Moderator Levels.

Term	nexp	ctrl	kcomp	kstudies	estimate	se	tval	df	pval	ci.lb	ci.ub
factor(inaction)action	70598	71676	306	51	0.0021344	0.0508705	0.0419581	344	0.9665565	-0.0979219	0.1021907
factor(inaction)inaction	4744	4170	40	23	-0.2874128	0.0691427	-4.1568044	344	0.0000408	-0.4234086	-0.1514171

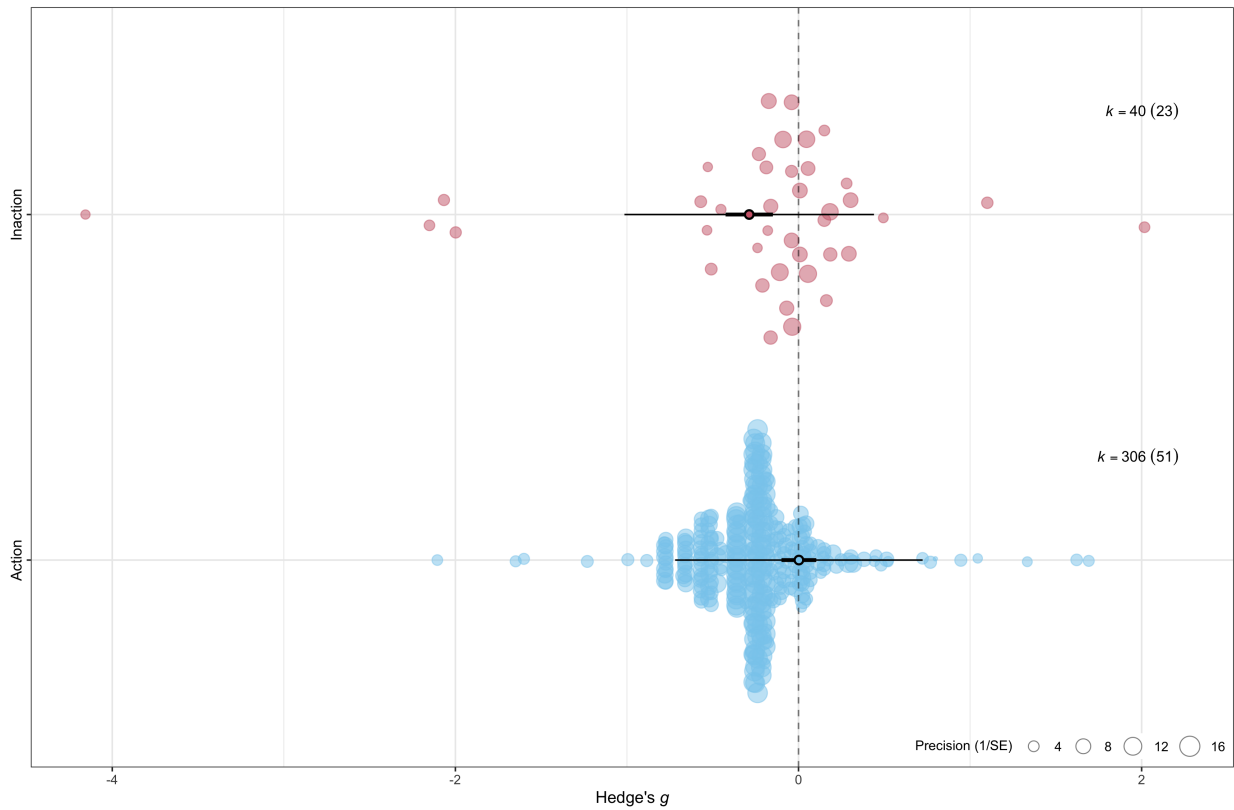
TOST results for each moderator level:

Action: $g = 0.00, 90\% \text{ CI } [-0.08, 0.09], \text{ bounds } \pm 0.15, p = 0.003$

Inaction: $g = -0.29, 90\% \text{ CI } [-0.40, -0.17], \text{ bounds } \pm 0.24, p = 0.768$

Figure S8

Orchard Plot of Decision Type Moderator



2.1.4 Responsibility

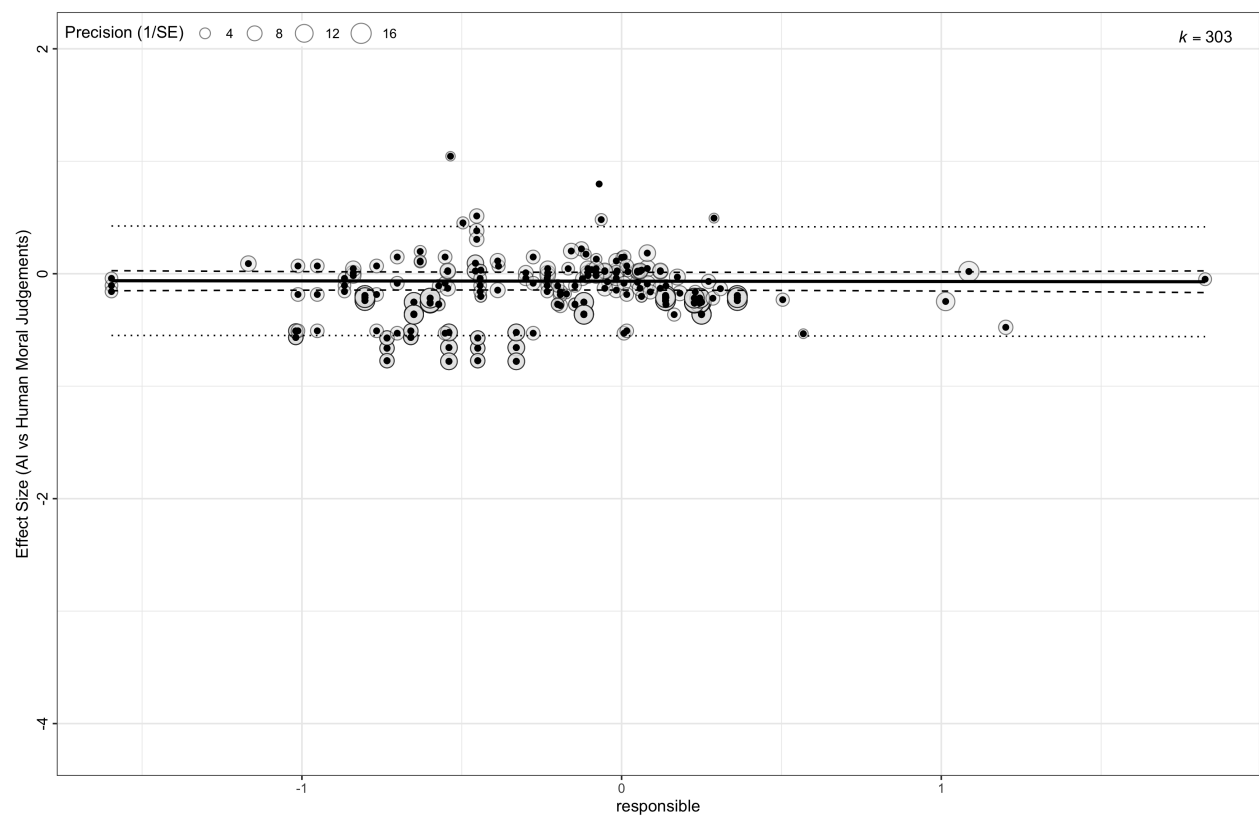
Moderation effect: $g = -0.00$, $p = 0.852$

Heterogeneity test: $Q_{(df=301)}=1190.79$, $p \leq .001$

TOST results: $g = -0.00$, 90% CI $[-0.03, 0.02]$, bounds ± 0.04 , $p = 0.004$

Figure S9

Orchard Plot of Responsibility Moderator



2.1.5 PMC

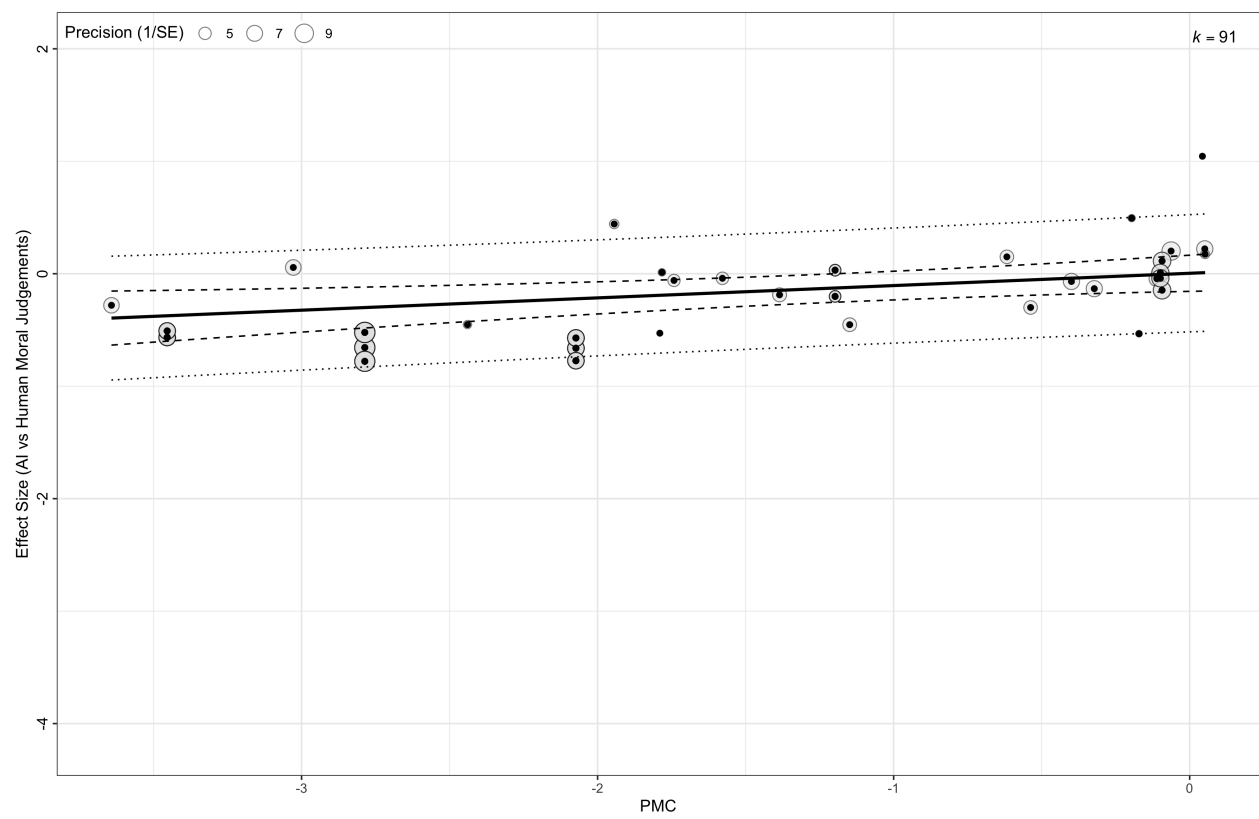
Moderation effect: $g = 0.11, p = 0.007$

Heterogeneity test: $Q_{(df=89)}=210.88, p \leq .001$

TOST results: $g = 0.11, 90\% \text{ CI } [0.04, 0.17], \text{ bounds } \pm 0.11, p = 0.483$

Figure S10

Orchard Plot of PMC Moderator



2.1.6 PMA

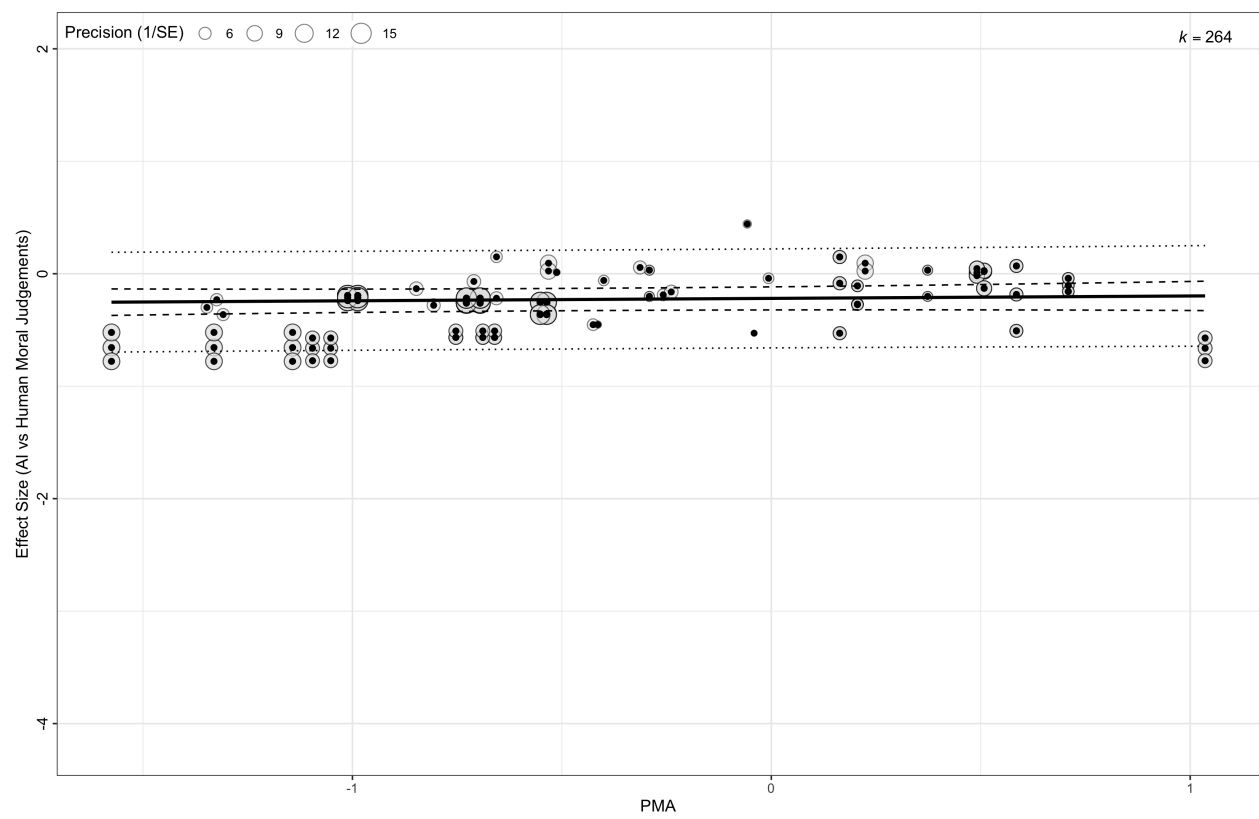
Moderation effect: $g = 0.02$, $p = 0.432$

Heterogeneity test: $Q_{(df=262)}=783.13$, $p \leq .001$

TOST results: $g = 0.02$, 90% CI $[-0.02, 0.07]$, bounds ± 0.08 , $p = 0.020$

Figure S11

Orchard Plot of PMA Moderator



2.1.7 Implied Intelligence of Agent (Implied vs. Not Implied)

Moderation effect: $F_{1,78}=0.99, p= 0.322$

Heterogeneity test: $Q_{(df=365)}=2230.25, p\leq .001$

Table S7

Effects of Each Implied Intelligence of Agent Moderator Levels.

Term	nexp	ctrl	kcomp	kstudies	estimate	se	tval	df	pval	ci.lb	ci.ub
factor(agentintel)implied	15230	14535	108	48	-0.0406387	0.0521123	-0.7798295	78	0.4378511	-0.1443863	0.0631089
factor(agentintel)notImplied	62723	63568	259	32	-0.1191807	0.0590980	-2.0166628	78	0.0471724	-0.2368357	-0.0015257

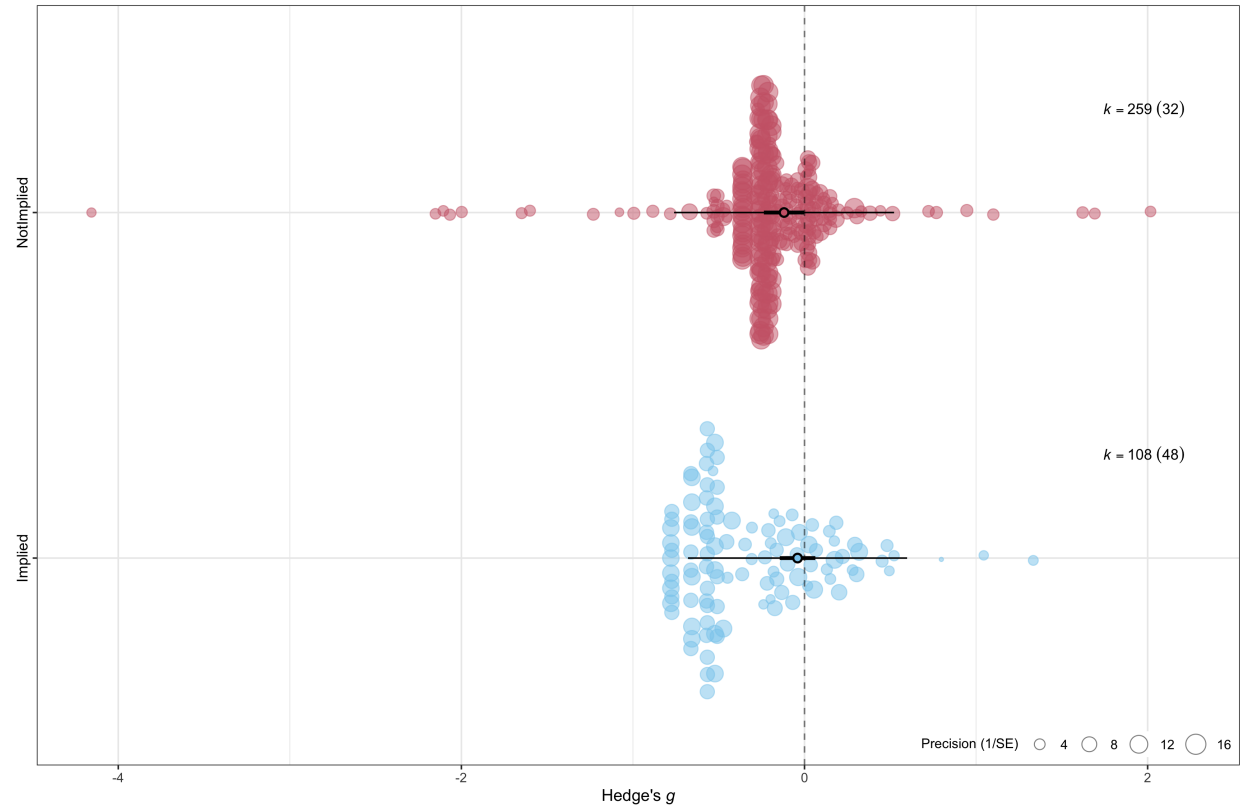
TOST results for each moderator level:

Implied: $g = -0.04$, 90% CI $[-0.13, 0.05]$, bounds ± 0.15 , $p = 0.021$

Not implied: $g = -0.12$, 90% CI $[-0.22, -0.02]$, bounds ± 0.17 , $p = 0.198$

Figure S12

Orchard Plot of Implied Agent Intelligence Moderator



2.1.8 Operationalised AI Categories

2.1.8.1 Two-Category Model (AI Systems vs. Robots)

Moderation effect: $F_{1,365}=0.40$, $p= 0.528$

Heterogeneity test: $Q_{(df=365)}=2020.49$, $p\leq .001$

Table S8

Effects of Each Two-Category AI Moderator Levels.

Term	nexp	ctrl	kcomp	kstudies	estimate	se	tval	df	pval	ci.lb	ci.ub
factor(aiTypea)AI	12885	12460	96	24	-0.1031168	0.0575991	-1.790250	365	0.0742424	-0.2163846	0.010151
factor(aiTypea)robot	65148	65723	271	57	-0.0642727	0.0429592	-1.496132	365	0.1354834	-0.1487513	0.020206

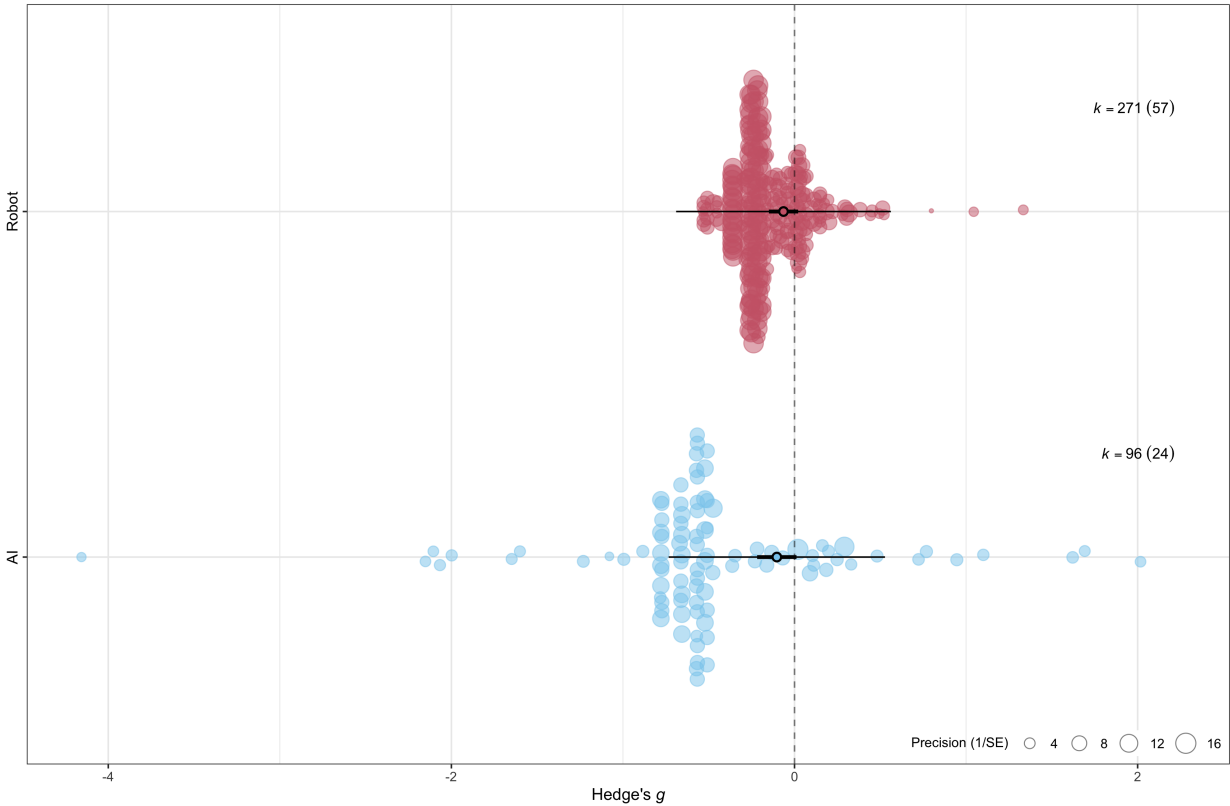
TOST results for each moderator level:

AI: $g = -0.10$, 90% CI $[-0.20, -0.01]$, bounds ± 0.20 , $p = 0.053$

Robot: $g = -0.06$, 90% CI $[-0.13, 0.01]$, bounds ± 0.13 , $p = 0.064$

Figure S13

Orchard Plot of Two-Category AI Moderator



2.1.8.2 Three-Category Model (AI Systems vs. Mechanical Robots vs. Humanoid Robots)

Moderation effect: $F_{2,364}=1.97$, $p= 0.141$

Heterogeneity test: $Q_{(df=364)}=1823.99$, $p\leq .001$

Table S9

Effects of Each Three-Category AI Moderator Levels.

Term	nexp	ctrl	kcomp	kstudies	estimate	se	tval	df	pval	ci.lb	ci.ub
factor(aiTypeb)AI	12885	12460	96	24	-0.0997394	0.0574738	-1.735388	364	0.0835184	-0.2127618	0.0132830
factor(aiTypeb)humanoid	52441	52826	163	15	-0.1117441	0.0497594	-2.245687	364	0.0253235	-0.2095961	-0.0138920
factor(aiTypeb)mechanical	12707	12897	108	51	-0.0535424	0.0431861	-1.239805	364	0.2158461	-0.1384680	0.0313832

TOST results for each moderator level:

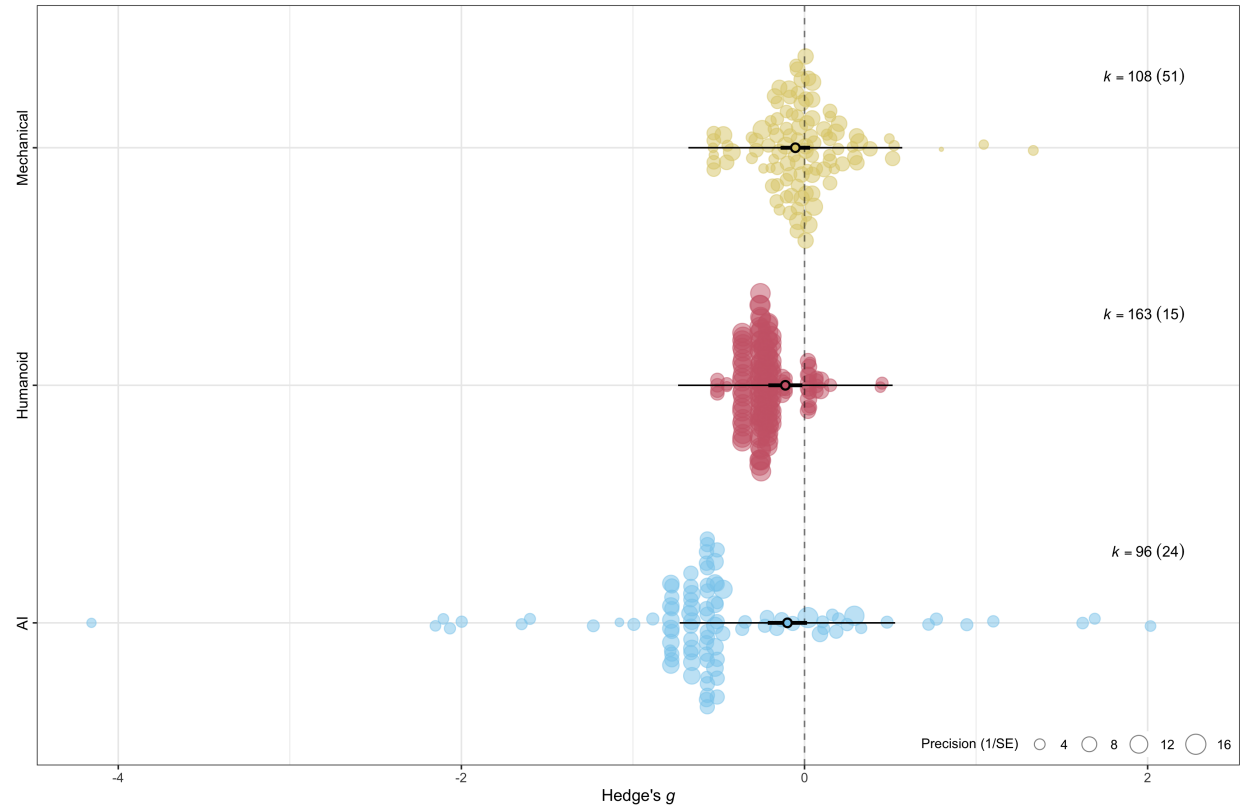
AI: $g = -0.10$, 90% CI $[-0.19, -0.01]$, bounds ± 0.20 , $p = 0.046$

Mechanical: $g = -0.05$, 90% CI $[-0.12, 0.02]$, bounds ± 0.14 , $p = 0.022$

Humanoid: $g = -0.11$, 90% CI $[-0.19, -0.03]$, bounds ± 0.25 , $p = 0.003$

Figure S14

Orchard Plot of Three-Category AI Moderator



2.1.9 Dependent Variable Wording

Moderation effect: $F_{8,71}=4.06$, $p < .001$

Heterogeneity test: $Q_{(df=358)}=2018.73$, $p \leq .001$

Table S10

Table showing effects of individual levels of DV wording moderator.

Level	k	Effect of Individual Level			
		g	95% CI		p
			Lower	Upper	
Acceptable	12	-0.13	-0.21	-0.04	0.003**
Appropriate	9	0.05	-0.10	0.20	0.474
Good	2	-0.01	-0.45	0.43	0.958
Justifiable	3	-0.34	-0.67	-0.02	0.038*
Moral	14	-0.14	-0.24	-0.05	0.004**
Permissible	26	-0.11	-0.19	-0.03	0.010**
Right	14	-0.04	-0.14	0.05	0.377
Unethical	4	-0.12	-0.22	-0.01	0.032*
Wrong	37	-0.07	-0.15	0.02	0.121

Note. * $p \leq .05$, ** $p \leq .01$, *** $p \leq .001$.

Table S11

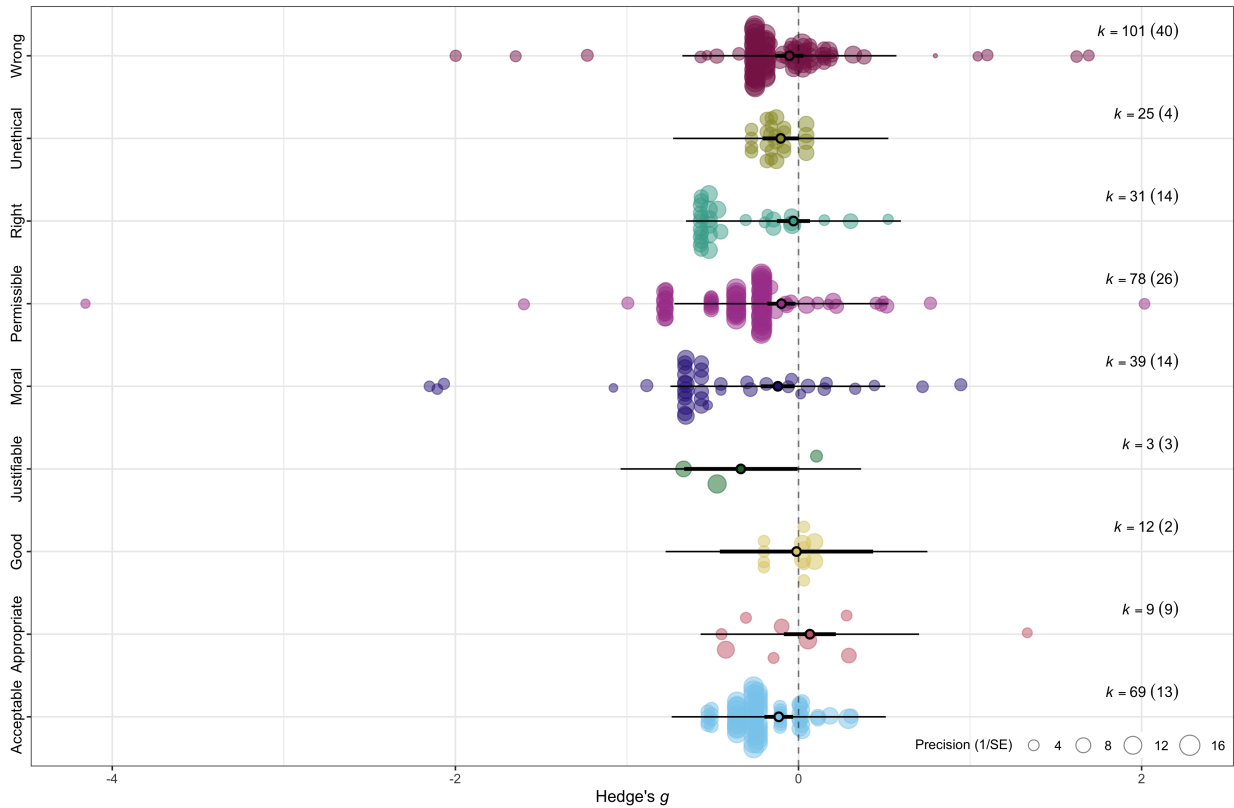
Table displaying the NHT results for each level of the DV wording moderator.

me	TOST					NHST p
	g	CI		Bounds ($\pm$)	p	
		Lower	Upper			
le	-0.13	-0.20	-0.06	0.26	0.001	0.003
ate	0.05	-0.07	0.18	0.35	0.000	0.474
	-0.01	-0.38	0.35	0.50	0.024	0.958
e	-0.34	-0.61	-0.07	0.57	0.080	0.038
	-0.14	-0.22	-0.06	0.26	0.008	0.004
le	-0.11	-0.18	-0.04	0.18	0.039	0.010
	-0.04	-0.12	0.04	0.26	0.000	0.377
l	-0.12	-0.20	-0.03	0.46	0.000	0.032
	-0.07	-0.13	0.00	0.16	0.012	0.121

Note. $p(-)$ and $p(+)$ correspond to p values of TOST at lower and upper bounds, respectively.

Figure S15

Orchard Plot of DV Wording Moderator



2.1.10 Research Quality

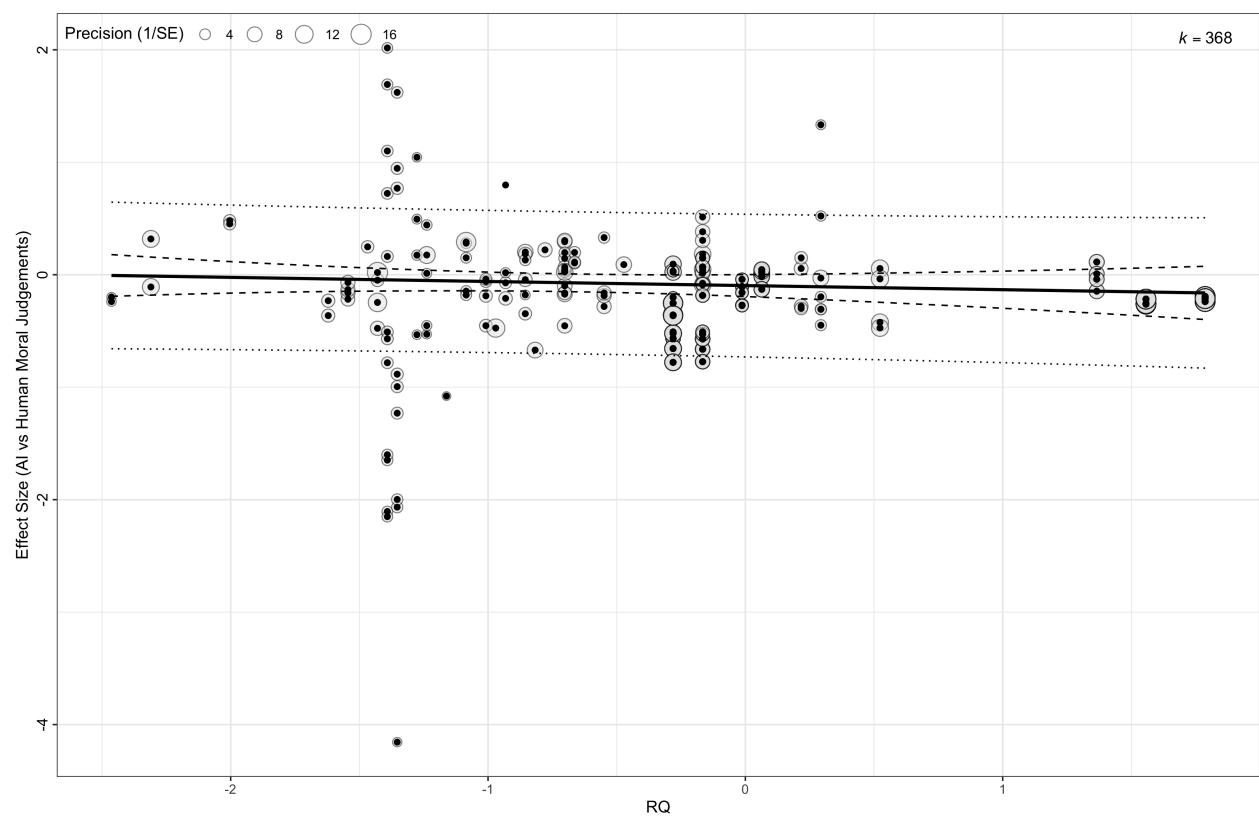
Moderation effect: $g = -0.04, p = 0.420$

Heterogeneity test: $Q_{(df=366)}=2404.97, p \leq .001$

TOST results: $g = -0.00, 90\% \text{ CI } [-0.01, 0.00], \text{ bounds } \pm 0.13, p = 0.000$

Figure S16

Orchard Plot of Research Quality Moderator



2.1.11 Multiple Regression Models

2.1.11.1 Intent and Decision Type As noted in the manuscript introduction, the literature suggests a complex relationship between the levels of these two moderators. Therefore, we conducted a further analysis with both moderators in the model.

Moderation effect: $F_{2,295}=12.56, p< .001$

Heterogeneity test: $Q_{(df=295)}=1722.63, p\leq .001$

The NHST results for all levels of the intent and decision type moderators in the multiple moderator model:

Intent:

Means-to-end: $n_{AI} = 1777, n_{Human} = 1534, k_{PIDs} = 11, k_{comp} = 21, g = -0.15, 95\% \text{ CI} = [-0.33, .03], p = .098$

Side-effect: $n_{AI} = 65878, n_{Human} = 65598, k_{PIDs} = 46, k_{comp} = 277, g = .07, 95\% \text{ CI} = [-0.06, .20], p = .274$

Decision Type:

Action: $n_{AI} = 64271, n_{Human} = 64345, k_{PIDs} = 40, k_{comp} = 269, g = .26, 95\% \text{ CI} = [.09, .44], p = .003$

Inaction: $n_{AI} = 3384, n_{Human} = 2787, k_{PIDs} = 18, k_{comp} = 29, g = -0.26, 95\% \text{ CI} = [-0.44, -0.09], p = .003$

The TOST results:

Intent

Means-to-end: ($g = 0.15, 90\% \text{ CI} [0.00, 0.30], \text{ bounds } \pm 0.16, p = 0.464$)

Side-effect: ($g = -0.07, 90\% \text{ CI} [-0.18, 0.04], \text{ bounds } \pm 0.16, p = 0.083$)

Decision Type

Action: ($g = 0.26, 90\% \text{ CI} [0.12, 0.41], \text{ bounds } \pm 0.15, p = 0.896$)

Inaction: ($g = -0.26, 90\% \text{ CI} [-0.41, -0.12], \text{ bounds } \pm 0.23, p = 0.643$)